

Consider the product below...

Who was this designed for? (**age**, **gender**, **income**?)

What **opportunity** in the **market** did this fill?

How does this design **appeal** to the people it was designed for?

Why did someone design this?



Product in a Tin

Think, Pair, Share:

What makes a good design? (function, creativity, user need, appeal).





Handmade
Five Stones
Game in a
Tin



Skin/Hair
care in a
Tin

Packaging & Branding examples



Felt Play Food
in a Tin



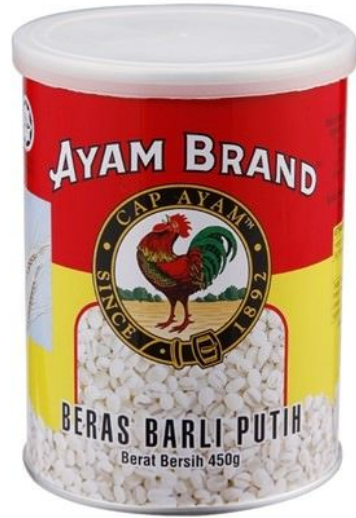
Fan in a Tin

Previous Student Examples:

Product Ideas:



Types of Tin Ideas:



Mind Map - AO1

- Understand the project brief and what makes a successful design outcome.
- Explore the creative potential of using tins as inspiration.
- Generate and record initial product ideas linked to a chosen tin.

1. What type of tin could you use?
2. What could fit inside?
3. Who might use it?
4. What problems could it solve?
5. What fun / novelty concepts could be popular?
6. What products reflect my interests, hobbies, or identity?
7. How could this tin become more than it looks?

Brand research

- brand sector: cosmetics (perfume)
- Logo: Dior
- Brand values: excellence & creativity, sustainability & responsibility, beauty & dreams
- Competitor brands: YSL, Gucci
- Customer base: teenage/ young women

Point of sale (POS)

- It will be displayed in cosmetic shops and Dior own stores
- Counter top display
- The POS should attract attention and hold the perfume as well as showing the overall theme and expressing the notes of the perfume
- The stand will have a slightly raised podium for the perfume to sit on and then will me surrounded by flowers on the bottom in a meadow like way, it will then also have the backdrop saying the name and brand. It will look like the wall with fairy lights making the perfume seem more apparent.

Sustainable & Ethics

- Eco friendly materials:
 - compostable card, recycled plastics, natural ingredients, soy wax
- Minimal packaging (no excess plastic)
- Honest labelling, will be made from natural ingredients and no animal testing
- Make POS displays out of recycled card and make it collapsible to limit packaging when transporting
- It can reflect my sustainable values by having a nature inspired design, be cruelty free and the perfume will be made of all natural ingredients.

Logo research

Fonts:

- Georgia
- Nicolas Cochon
- Kiss-Do
- Allusion, Papyrus
- Olvera

Product design

- My product will serum, body spa
- The function w
- Ergonomics: it will hold/designs hands) with a easily click into
- A customer who travel and app
- A customer who product as it o
- soft and floral luxury in a single

Personal identity and interests

- My hobbies/ passions: sewing, art, fashion, music
- Brands i already connect with: Brandy melville, dior, hollister, bath and body works, YSL, marc jacobs.
- Aesthetics that reflect me: cottage core, cottage, downtown, clean aesthetics, spring
- Causes i care about: climate change, anti animal testing, plastic pollution.
- my project will reflect my values by using recycled materials, using renewable energy and will not test on animals, use natural ingredients

**two locations: theme park w/
live concert & from trucks or sandy
beach w/ live concert and a chance
to learn how to surf**

**having varieties of food
so that people
won't be allergic
to them**

**all food and
drinks cost
above \$5 as
the money
goes to charity**

**will sell fruits as well as
ice cream as I will buy hot
the design will be aesthetically
and Instagrammable**

**outside,
as it will
take place
during the
summer**

**will also include
vegan
food (vegan
ice cream etc)**

**can't eat/smoothies/
chips/drinks
(no alcohol influencers)**

fashion festival

By Milia, 10.3

**influencers aged
13-21 can promote
this event via insta**

**clothing are
free as they
are sponsored
by the brand**

TASK

Use the questions provided to create a visual and aesthetic mind map that explores all of your ideas for the Product in a Tiny project.

How would you
this **situation**



to **design**...?



EXAMPLE



Drink tins, milk tins, old cans. Tins that have lids.



What type of tin?

How can this tin become more than it looks?

Product in a Tin

What product reflects me?

What could fit inside?

Who is it for?

What problems could it solve?

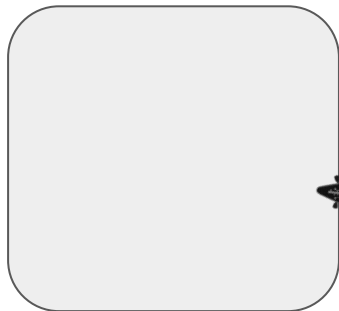
Fun concepts

EXAMPLE

Mindmap



How could this tin become more than it looks?



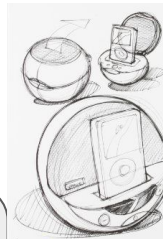
what products reflect me?



could this be environmental or sustainable?



What problems could it solve?



Product sketches and Ideas



Who is it for?



Fun Concepts

what could fit inside the tin?



what kind of tin could I use?



MIND MAP

Did I...

	Answer all 7 questions in my mind map
	Use a good mix of text responses and images
	Include a variety of ideas and answers
	Ensure that my mind map follows an aesthetic
	Upload my mind map as a slide in my DT Portfolio on Gclass

Learning Objectives:

- Write a clear design brief.
- Define a realistic target user.
- Identify user needs that influence design.

What is the **Situation**?

什么是情境 (Situation)?

상황(Situation)이란 무엇입니까?

The *situation* is the **background story** to the project.

情境是项目的背景说明。

상황은 프로젝트의 배경 이야기입니다.

It explains:

- **Why** the design is needed.
 - 为什么需要这个设计
 - 왜 이 디자인이 필요한지.
- **Who** might use it.
 - 谁可能会使用它
 - 누가 사용할 수 있는지.
- What **problem** or **opportunity** exists.
 - 存在什么问题或机会
 - 어떤 문제나 기회가 있는지.

Situation:

"I want to make my tin look cool."

Design Brief:

"I will design a tin for my friends."

Think, Pair, Share...

Are these good examples of Situation and Design Brief? Why?

Situation:

"Tins are fun."

Design Brief:

"I will rebrand a tin and put random stuff inside."

Design Brief & Customer Profile - AO1

What is the **Design Brief**?

什么是设计任务 (Design Brief)?

디자인 과제(Design Brief)란 무엇입니까?

The *design brief* is a **short statement of intent**.

设计任务是一段简短的意图说明。

디자인 과제는 간단한 의도 진술문입니다.

It explains:

- **What** you are going to **design**.
 - 你要设计什么
 - 무엇을 디자인할 것인지.
- **Who** you are designing for.
 - 你是为谁设计的
 - 누구를 위해 디자인하는지.
- The **purpose** of the **product**.
 - 产品的目的是什么
 - 제품의 목적이 무엇인지.

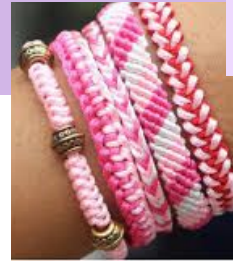
Learning Objectives:

- Write a clear design brief.
- Define a realistic target user.
- Identify user needs that influence design.

Design Brief & Customer Profile - AO1

TASK 1: Research & Investigate

1. In your **digital portfolio**, write a **situation** (background story) explaining why your design is needed.
2. Write a **design brief** (mission statement) explaining what you will design and who it is for.



Sentence Starters to Help You

- ### 1) Situation starter:

"People often struggle with _____. A recycled tin could be reused as _____ to help _____."

- ## 2) Design Brief starter:

"I will design and make a _____ that fits inside a recycled tin. The tin will be rebranded to appeal to _____."

Learning Objectives:

- Write a clear design brief.
- Define a realistic target user.
- Identify user needs that influence design.

Design Brief & Customer Profile - AO1

SELF CHECK:

Tick if your **Situation** and **Design Brief**:

- ☐ Clearly explain the problem or need
- ☐ Identify a user or customer
- ☐ Say what product will go inside the tin
- ☐ Describe how the tin will be rebranded
- ☐ Are realistic and can be made in school & at home

Research & Investigate

1. Write a **situation** (background story) explaining why your design is needed.
2. Write a **design brief** (mission statement) explaining what you will design and who it is for.



Sentence Starters to Help You

Situation starter:

"People often struggle with _____. A recycled tin could be reused as _____ to help _____."

Design Brief starter:

"I will design and make a _____ that fits inside a recycled tin. The tin will be rebranded to appeal to _____."

Pocket-Sized Puzzle Game

Situation:

"Teenagers often look for quick, fun activities to pass the time while waiting for friends or during breaks. Many games are too bulky to carry around. A recycled tin could be reused to hold a compact puzzle or game."

Design Brief:

"I will design and make a mini puzzle game that fits inside a recycled tin. The rebranding will give the tin a playful, bold style to appeal to teenagers who enjoy portable entertainment."

Mini Craft Kit

Situation:

"Creative hobbies like bracelet making or origami are popular with young people, but materials can get lost or damaged without proper storage. A recycled tin could provide a portable craft kit."

Design Brief:

"I will design and make a craft kit inside a recycled tin, such as a mini friendship bracelet set. The rebranding will target creative teenagers who enjoy DIY projects."

Winter Accessories Tin

Situation:

"Small clothing items like gloves, socks, or hats are often misplaced. Repurposing a tin as packaging could keep items together while promoting a reusable storage solution."

Design Brief:

"I will design and make a pair of hand-knitted socks to go inside a recycled tin. The rebranding will give the product a modern, eco-conscious look to attract environmentally aware teenagers."

Pocket First Aid / Care Kit (Non-medical, Product Focus)

Situation:

"Young people often need small useful items like plasters, lip balm, or hand wipes when on the go, but they can get lost at the bottom of a bag. A recycled tin could be repurposed to hold these essentials."

Design Brief:

"I will design and make a small personal care kit inside a recycled tin, including a simple hand-made item like reusable wipes. The rebranding will emphasise health and sustainability for active teenagers."

SITUATION AND DESIGN BRIEF

Did I...

	Clearly explain the problem or need
	Identify a user or customer
	Say what product will go inside the tin
	Describe how the tin will be rebranded
	Ensure my situation and design brief are realistic and can be made in school & at home

Consider the **3 types of products** below.
Who do you think they were made for? Describe that **demographic**



Learning Objectives:

- Write a clear design brief.
- Define a realistic target user.
- Identify user needs that influence design.

Design Brief & Customer Profile - AO1

- Every product is designed for someone in particular
- A customer profile helps you understand the customer's needs before designing

Understand what the customer needs/wants

a better designed product



Demographic 人口统计 인구통계학적

a way to describe **groups of people** by **things** they have in **common**, such as **age**, **gender**, **where they live**, or **jobs**.

CUSTOMER PROFILE

Demographics

who they are

- Factor One
- Factor Two
- Factor Three
- Factor Four

- Factor Five
- Factor Six
- Factor Seven

Psychographics

how they think

- Factor One
- Factor Two
- Factor Three
- Factor Four

- Factor Five
- Factor Six
- Factor Seven

Buying Behaviors

what they do

- Factor One
- Factor Two
- Factor Three
- Factor Four

- Factor Five
- Factor Six
- Factor Seven

Desires

what they want

- Factor One
- Factor Two
- Factor Three
- Factor Four

- Factor Five
- Factor Six
- Factor Seven

Learning Objectives:

- Write a clear design brief.
- Define a realistic target user.
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Design Brief & Customer Profile - AO1

TASK:

- Who is your customer?
- What age are they?
- What are their interests?
- What are their needs? (family, pets, lifestyle etc)
- What is their job/income?
- Do they have any desires or specific requirements?



Extension

What are other questions will help you understand the customer needs?
What similarities or differences do you have with your partner? Compare!

EXIT TICKET QUESTIONS:

Why is a mindmap helpful at the beginning of a project?

What does a customer profile help a designer do?

Who is your customer for the Product in a Tin?

Reflect on how your group collaboration is going so far.

EXAMPLE



Who They Are

- Female
- Age 24
- Works at a music shop in Damansara
- Lives alone
- Has 2 pet cats
- Plays the electric guitar
- Loves drinking coffee

What They Think

- Being environment friendly is very important
- Art and music plays a big part in her life
- Expensive luxury brands are a waste of money

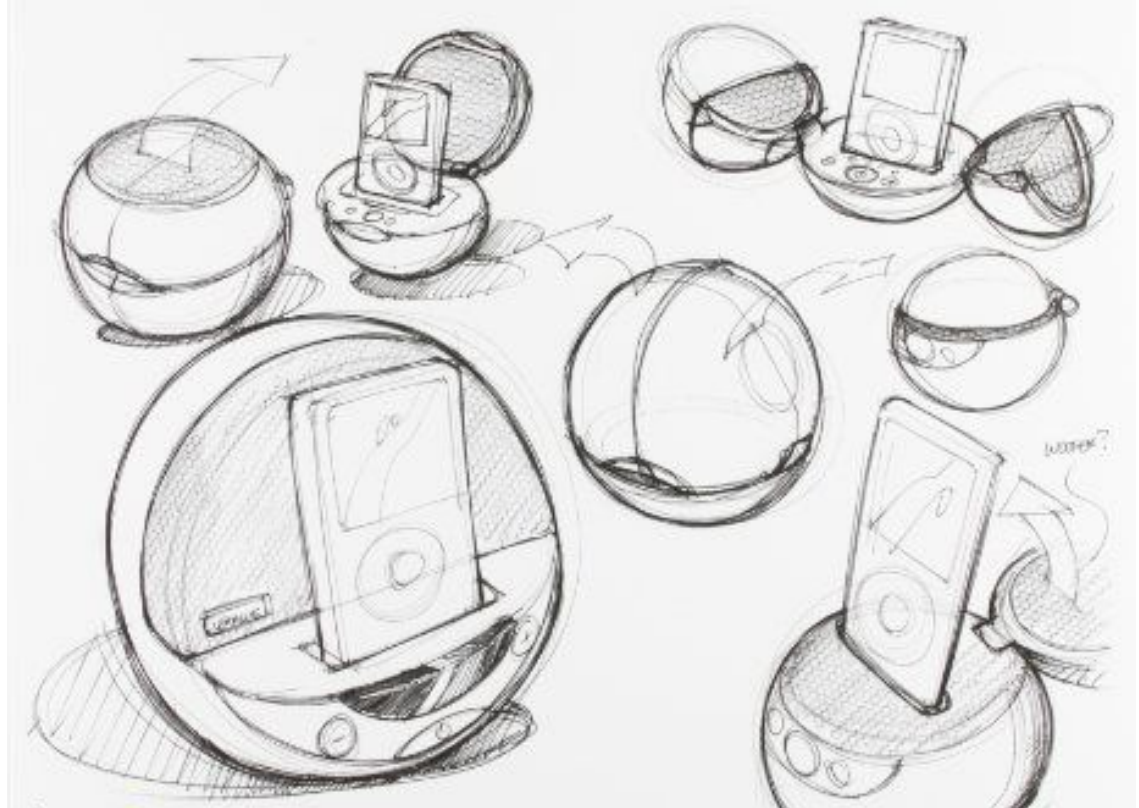
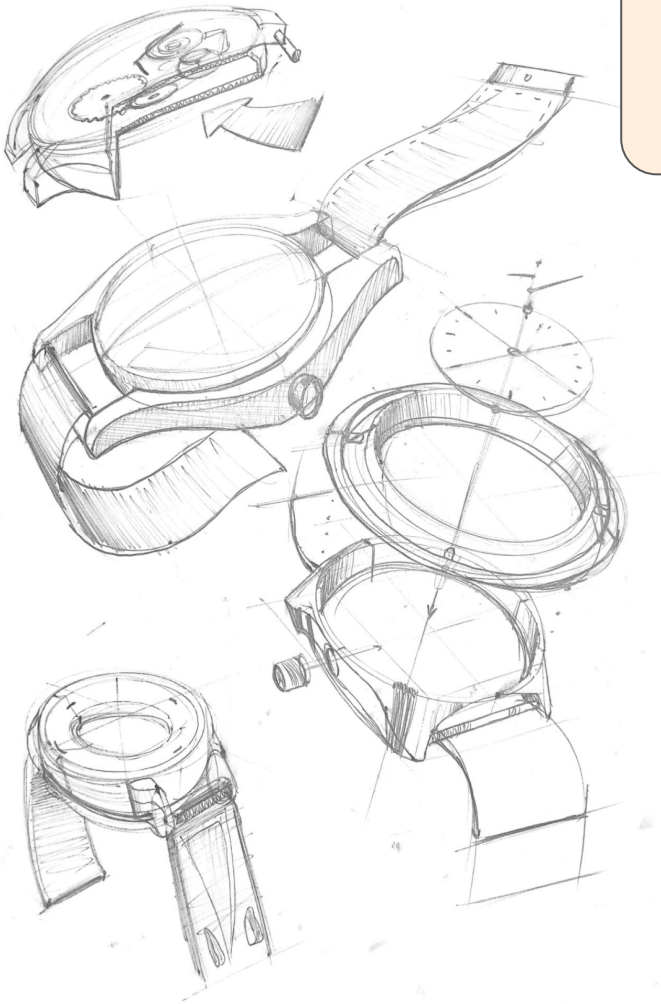
What They Do

- Teaches music part time
- Is a part of a band and plays at various different venues late at night
- Takes her cats to cafes on the weekends

What They Want

- A modern and chic product with dark colours
- Something portable and easy to carry
- Something useful and aesthetic

These are examples of product design sketches.
What do you notice about these drawings?
What is the benefit of making drawings like these?

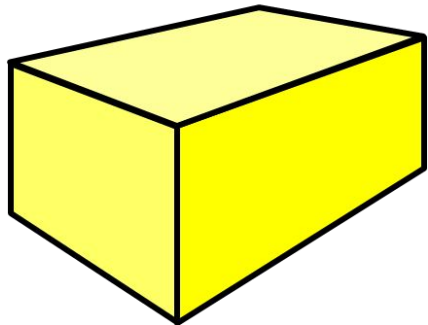
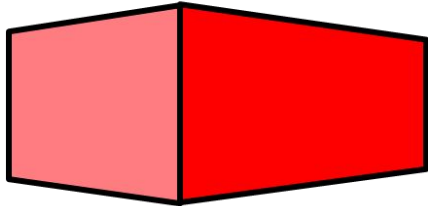
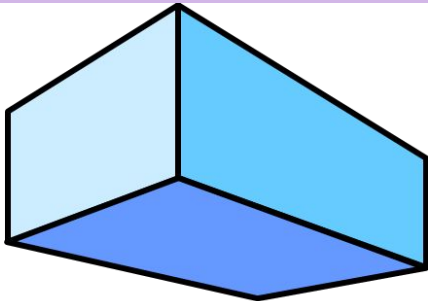


Learning Objectives:

- To be able to draw in 2 point perspective and add appropriate colour.
- To develop various shape designs using 2 point perspective skills.

2 Point Perspective

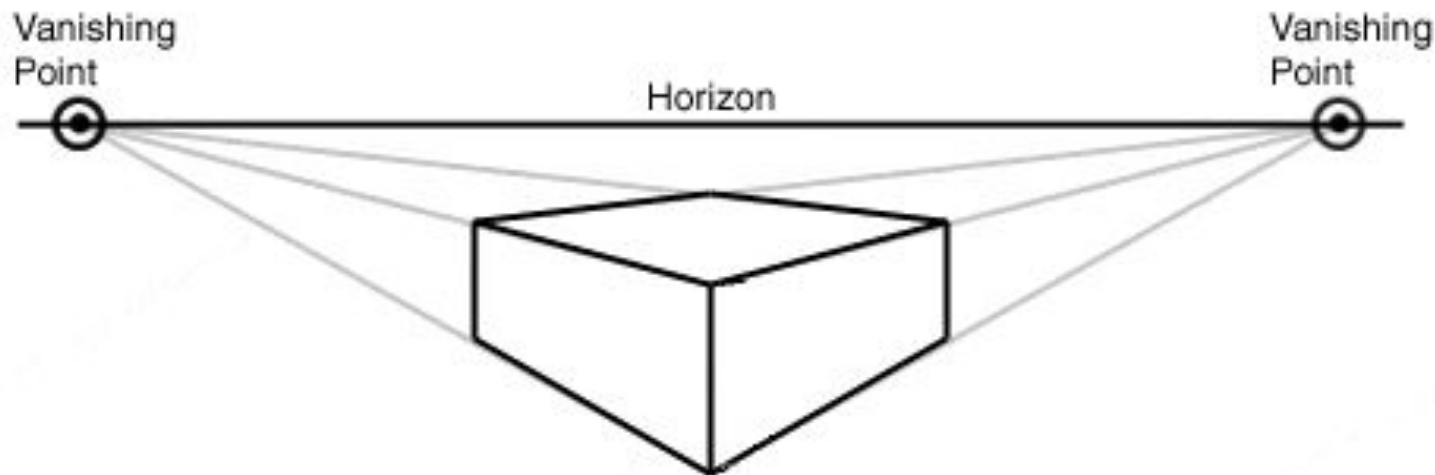
**What is
perspective?**



English	Mandarin	Korean
Perspective	透視 (tòu shì)	원근법 (won-geun-beop)
Horizon	地平线 (dì píng xiàn)	지평선 (ji-pyeong-seon)
Vanishing Point	消失点 (xiāo shī diǎn)	소실점 (so-sil-jeom)
Cuboid	长方体 (cháng fāng tǐ)	직육면체 (jik-yuk-myeon-che)
Cylinder	圆柱体 (yuán zhù tǐ)	원기둥 (won-gi-dung)

Learning Objectives:

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- To develop various shape designs using 2 point perspective skills.



- Horizon
- **Two** vanishing points
- draw from the corner
- Easy way to add depth to a design
- Visual tool to communicate an idea

2 Point Perspective

A drawing method where objects are shown with two vanishing points on the horizon line. This creates a sense of depth, showing two sides of an object receding into the distance.

两点透视

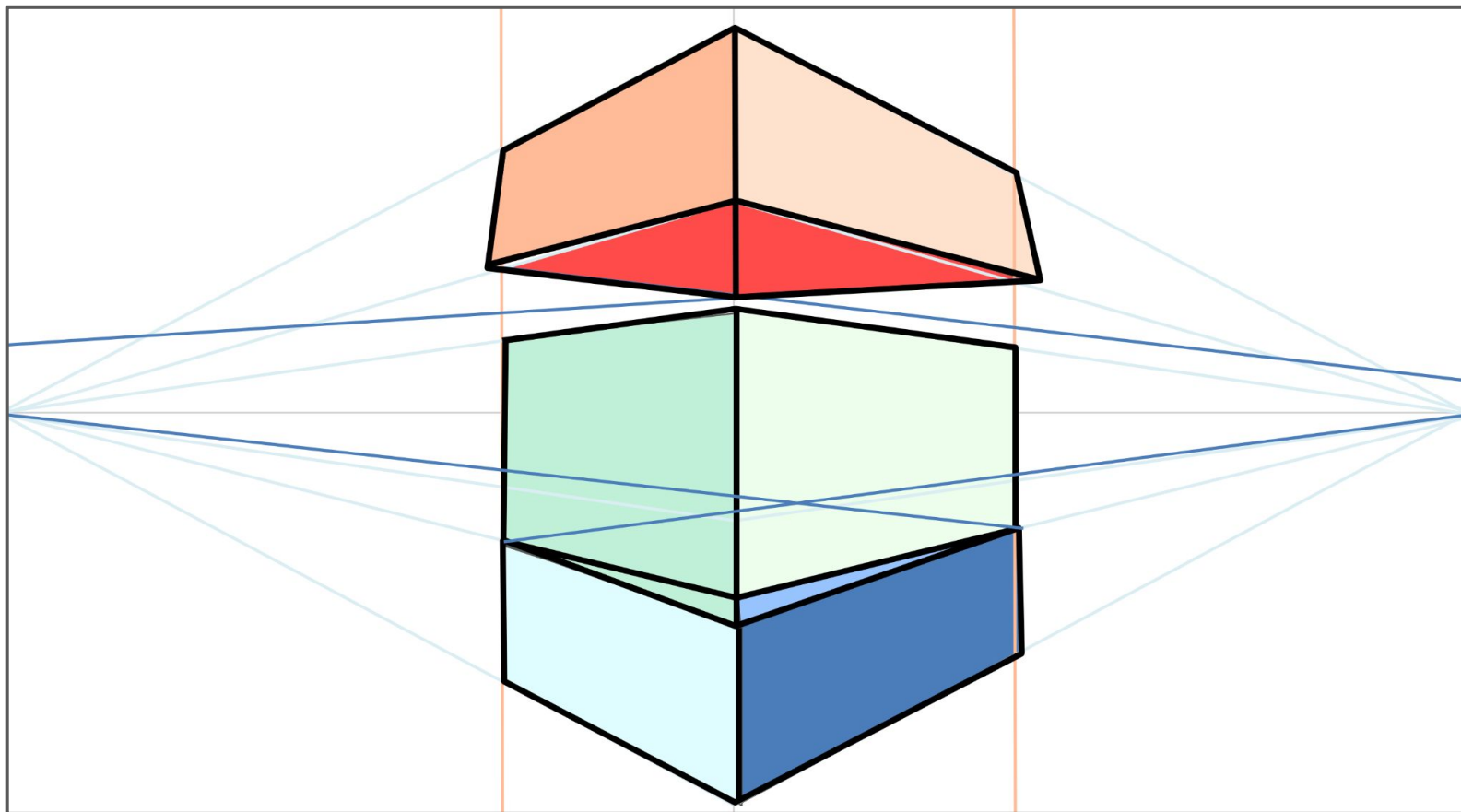
一种绘画方法，在地平线上有两个消失点，使物体的两侧向远处消失，产生空间和深度的感觉

투점 원근법

지평선 위에 두 개의 소실점을 사용하여 물체의 두 면이 멀어지며 사라지는 것처럼 그려 공간감과 깊이를 표현하는 기법.



What **mistakes** can you spot in this **2 point perspective**?
Use the **key words** to describe what you see!



Key Words

Parallel

Horizon

Vanishing
point

Cuboid

Lines

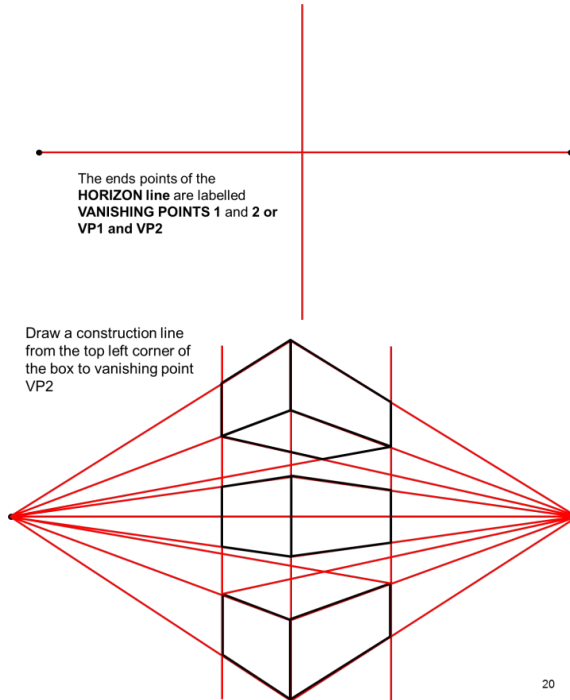
Shading

Light source

Learning Objectives:

- To be able to draw in 2 point perspective and add appropriate colour.
- To develop various shape designs using 2 point perspective skills.

Task 1: Using the hand outs for help, create a Birds eye view, Eye level view and Worm's eye view cuboid.



20

Add weight to the lines shown to create your first block this is known as the **EYE LEVEL VIEW**

This diagram shows the next step in the construction of the cuboid. The lines are drawn with more weight to create a three-dimensional effect. The cuboid is shown in a perspective view, with its top and bottom edges receding towards the vanishing points. The text indicates that this is the 'EYE LEVEL VIEW'.

With the light coming from the left add colour and shade to the blocks

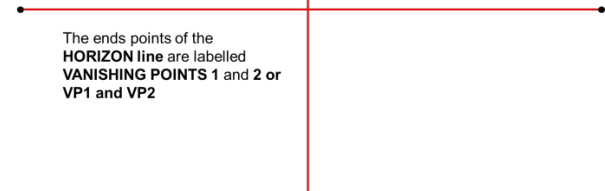
This diagram shows the final step in the construction of the cuboid. The cuboid is colored and shaded to represent light coming from the left. The top face is light blue, the front face is dark blue, and the bottom face is yellow. The text indicates that this is the 'EYE LEVEL VIEW'.

Draw a construction line from the bottom right corner of the box to vanishing point **VP1**

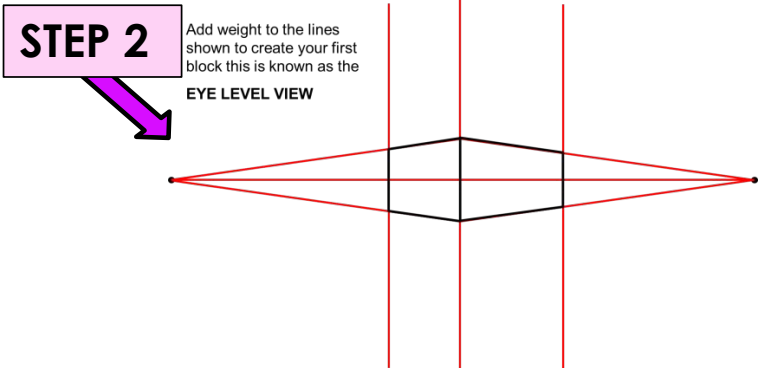
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Peer assess, does your shoulder buddy understand what birds eye, worms eye and level view are?
Have they demonstrated this in their work?

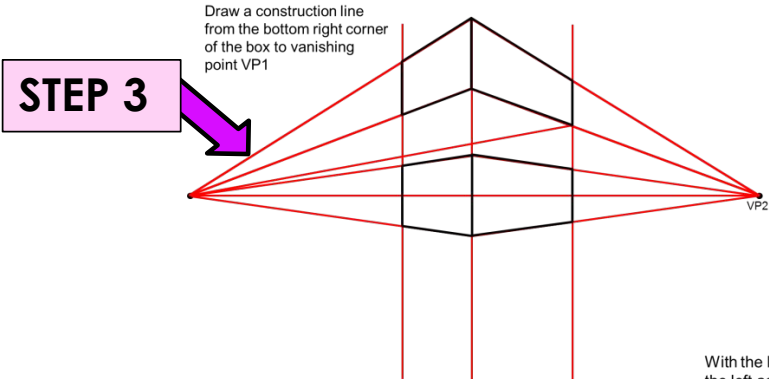
STEP 1



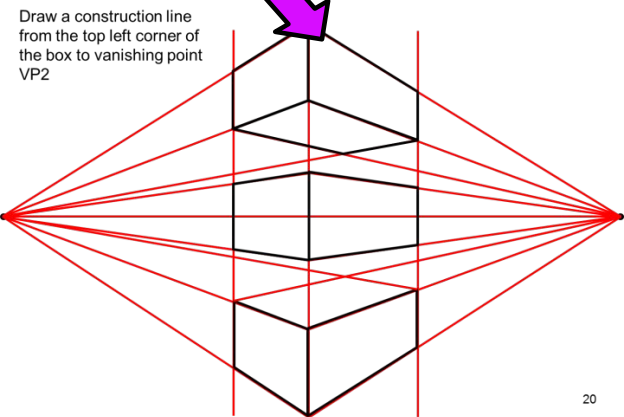
STEP 2



STEP 3

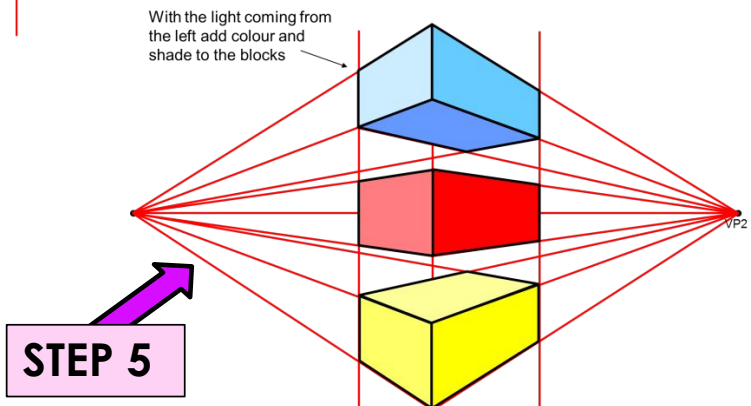


STEP 4



2-Point Perspective Help Sheet

STEP 5

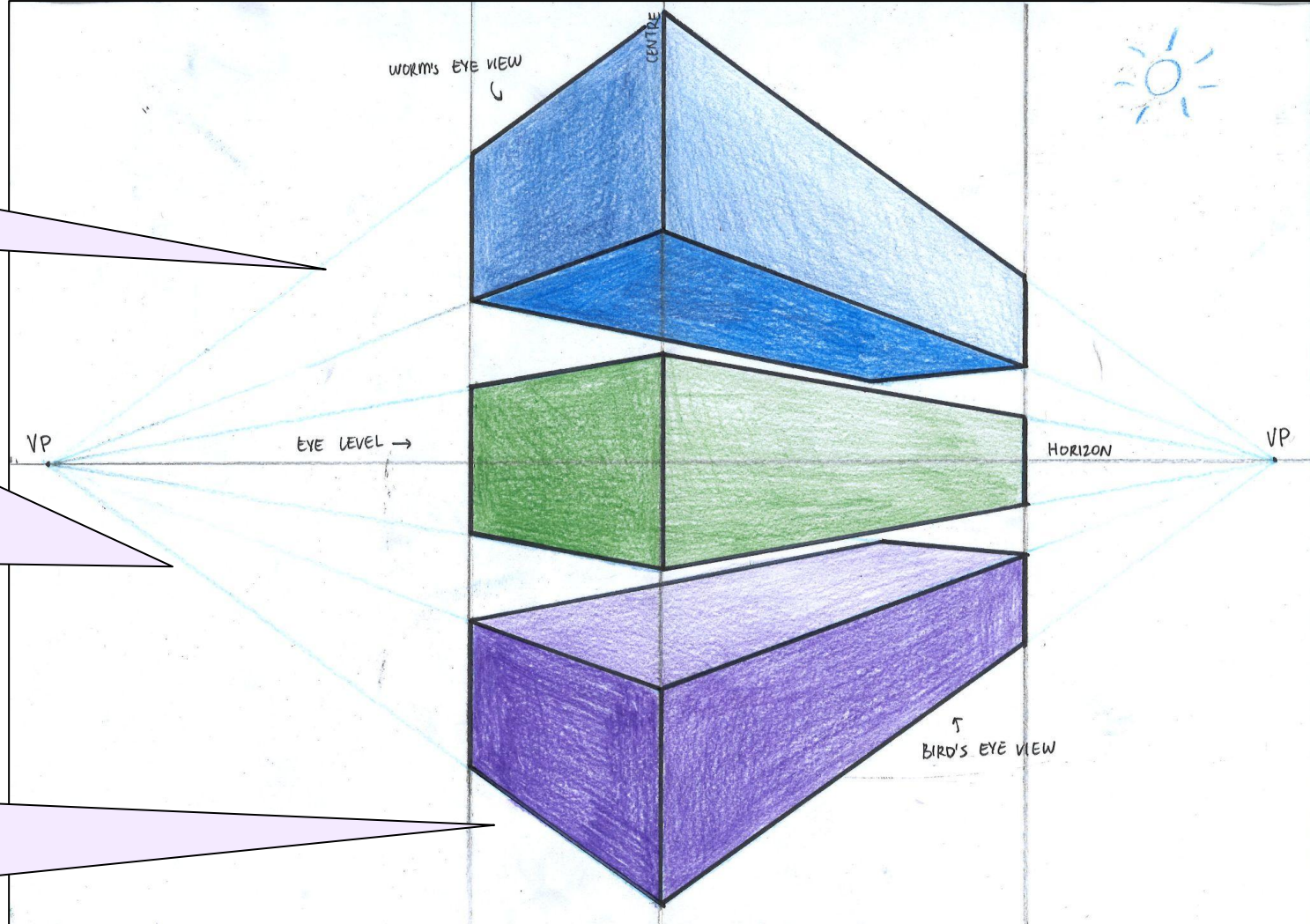


EXAMPLE WORK

Working lines and **labels** are clearly seen

2 point perspective shows **3 views** (**bird's, eye level, worm's**) and uses the right **techniques**

Cuboids are rendered with a **light, medium and dark face**, showing a consistent **light source**

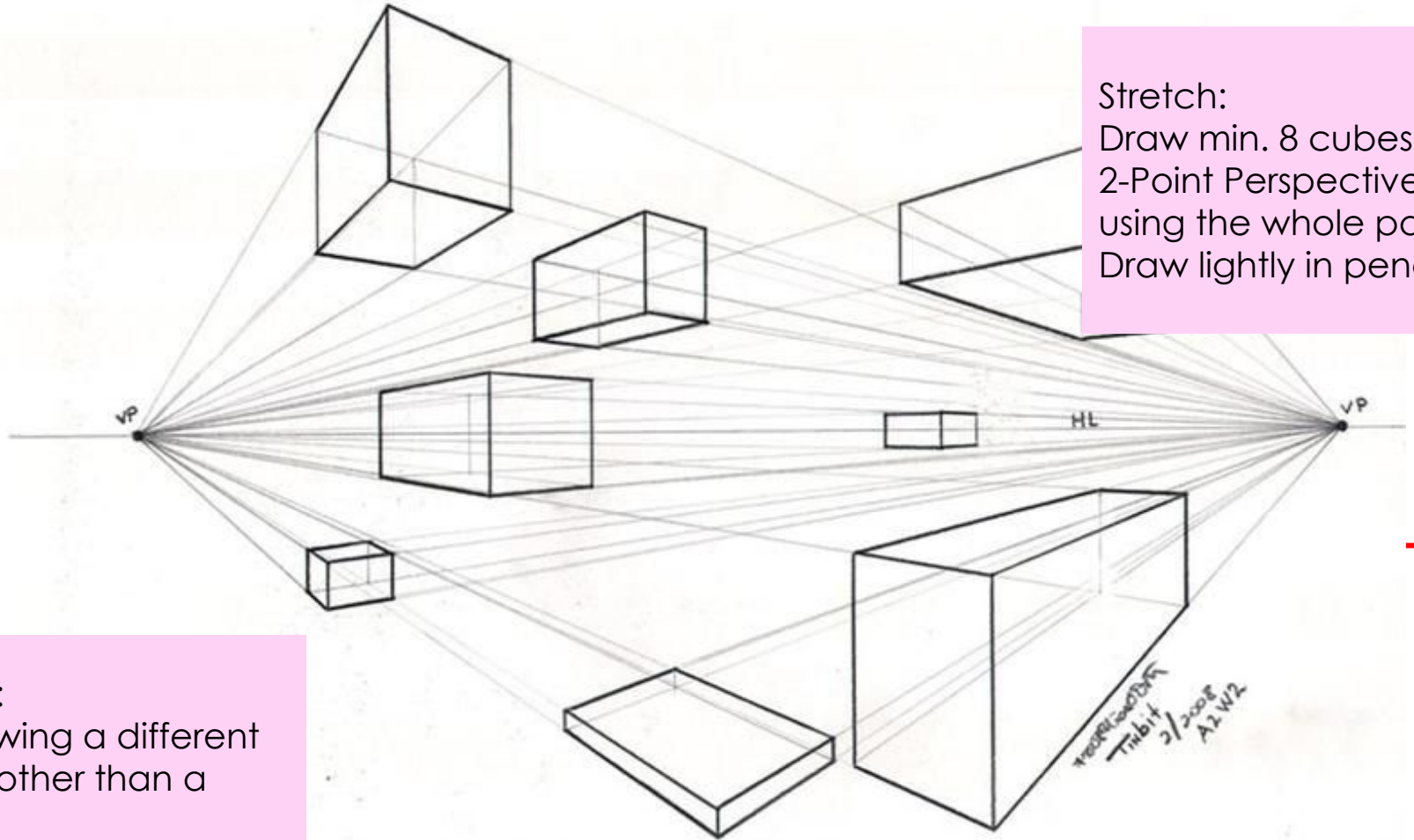


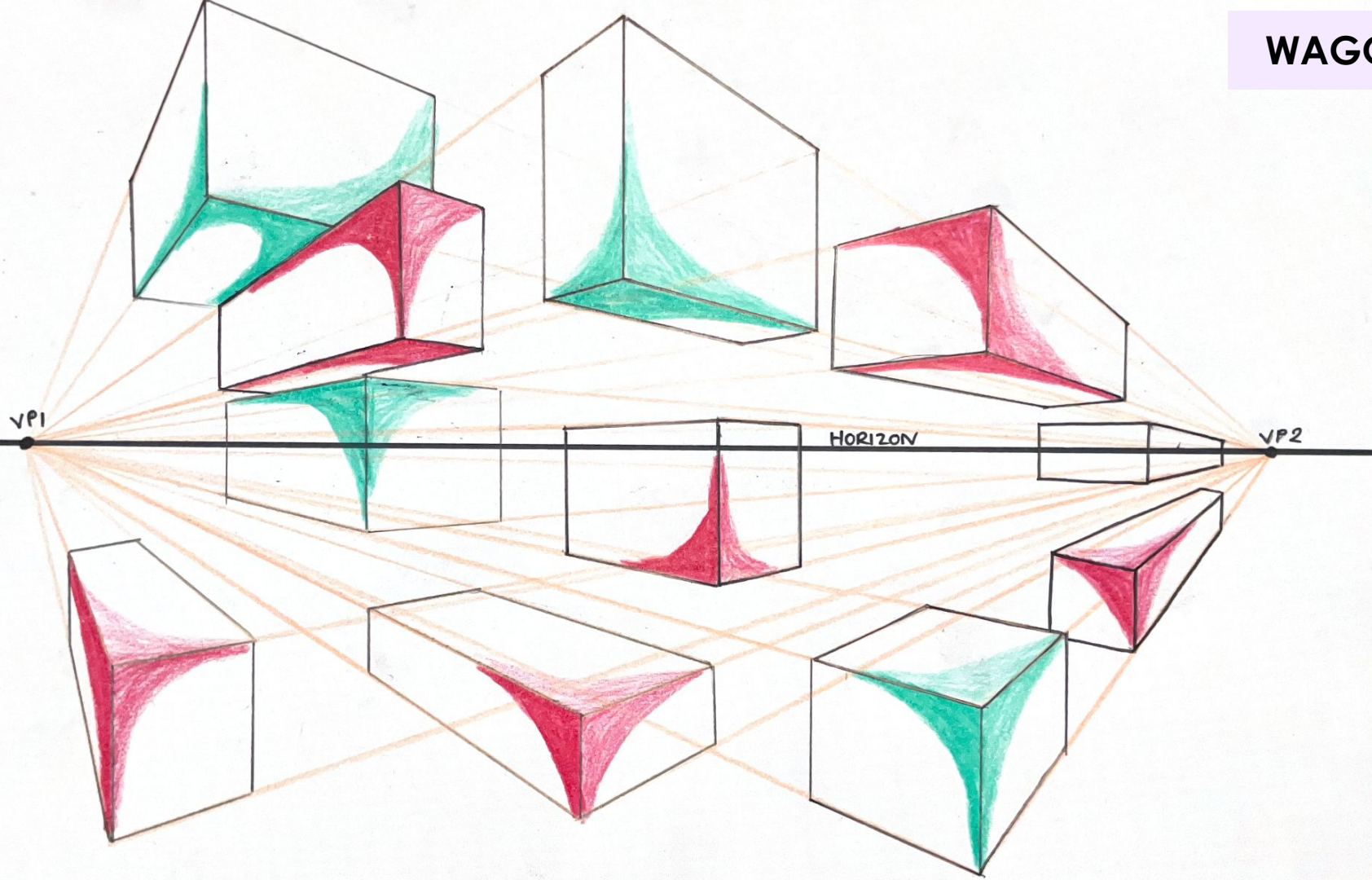
Stretch:
Draw min. 8 cubes in
2-Point Perspective
using the whole page.
Draw lightly in pencil.

VP

VP

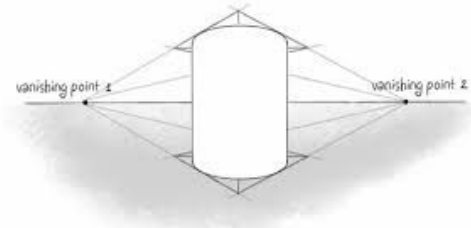
Further:
Try drawing a different
shape other than a
cube.



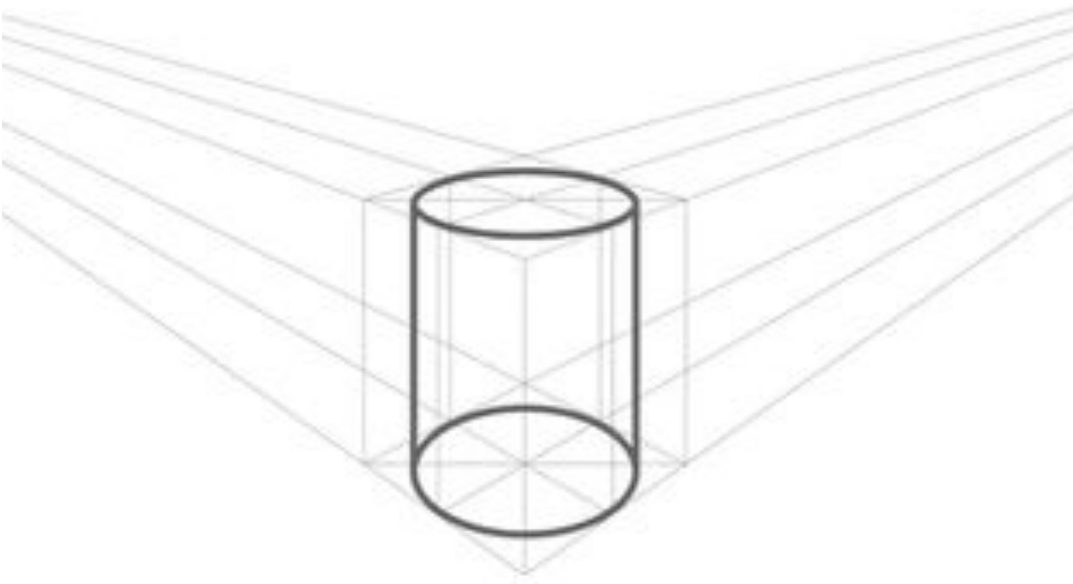


Learning Objectives:

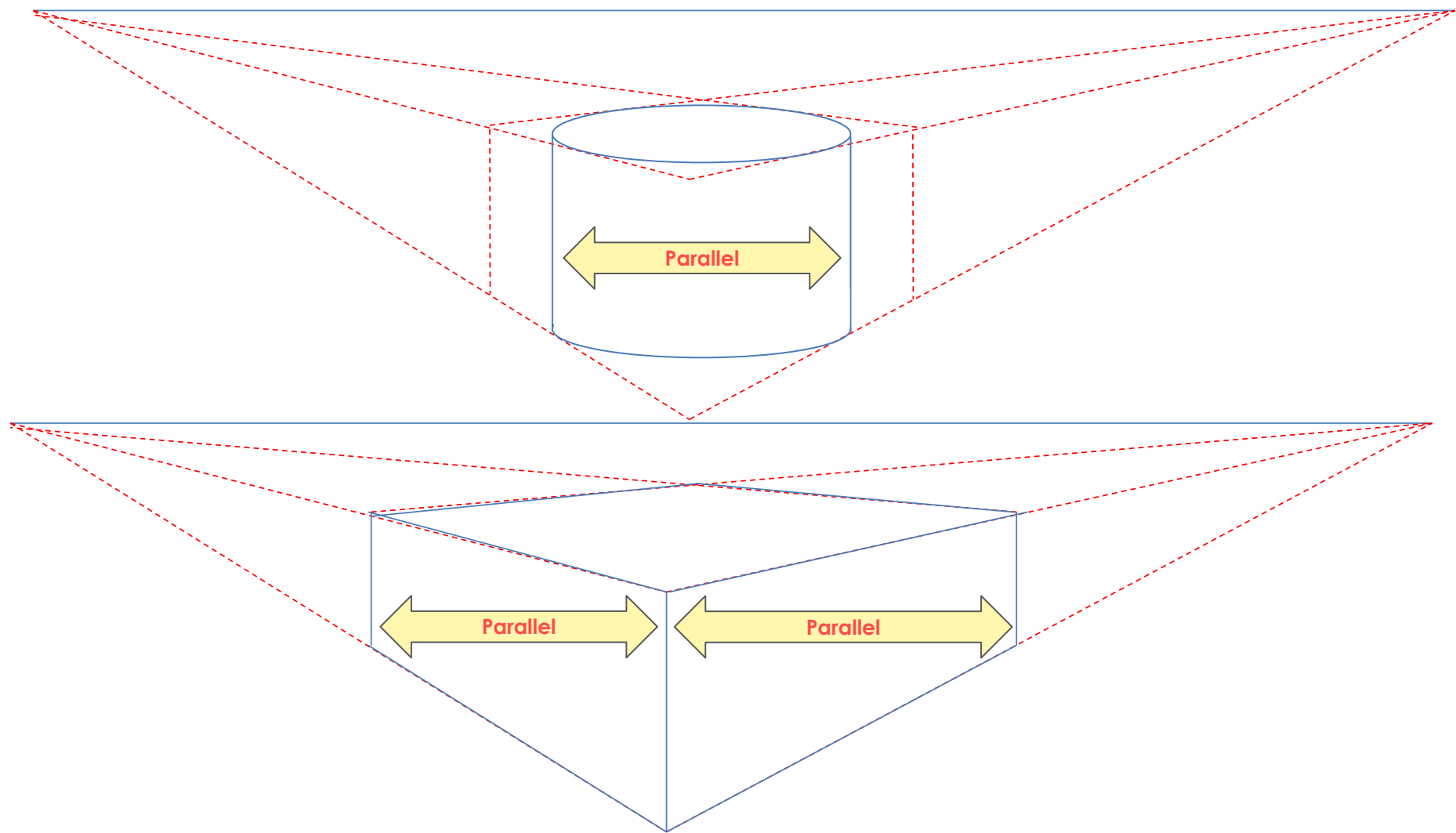
- To be able to draw in 2 point perspective and add appropriate colour.
- To develop various shape designs using 2 point perspective skills.

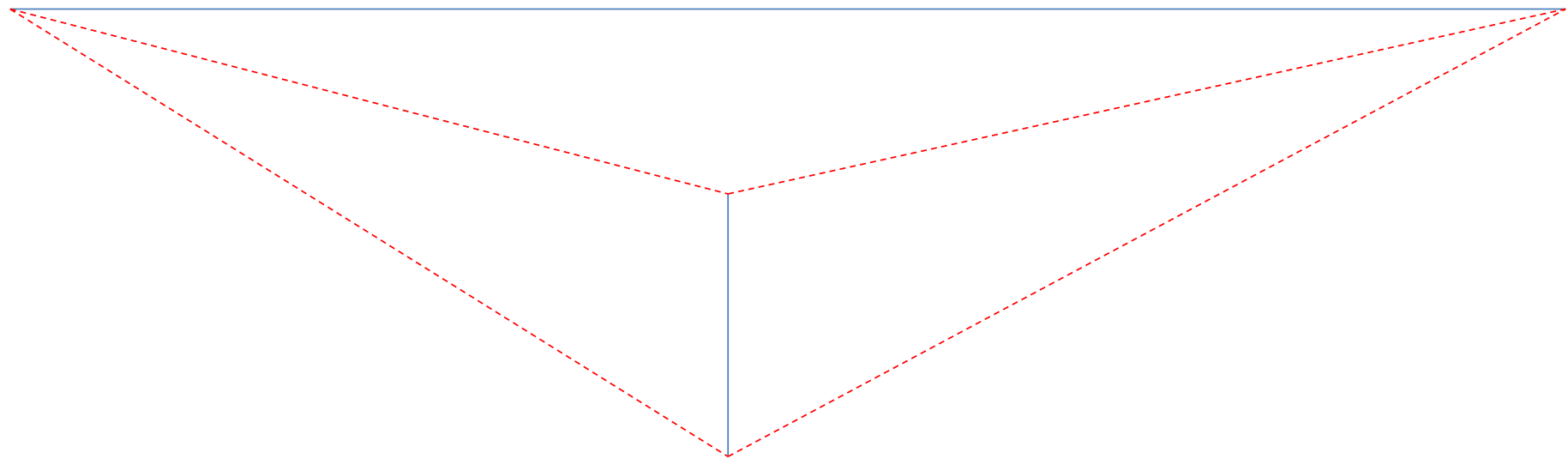


Task 1:
Using the skills you have just developed you must complete 6 initial ideas for your tin design, consider the shape of your chosen tin when using your 2 point perspective skills.



Assessment	
STRETCH	My Final design has the following:- • 2 Point Perspective Drawing • Good use of Shading • Good use of Rendering • 2 tone shading • Annotation
CHALLENGE	My Final design has the following:- • 2 Point Perspective Drawing • Use of Shading • Use of Rendering • Notes about your designs
EXPECTED	My Final design has the following:- • 2 Point Perspective Drawing • Poor use of Shading • Poor use of Rendering





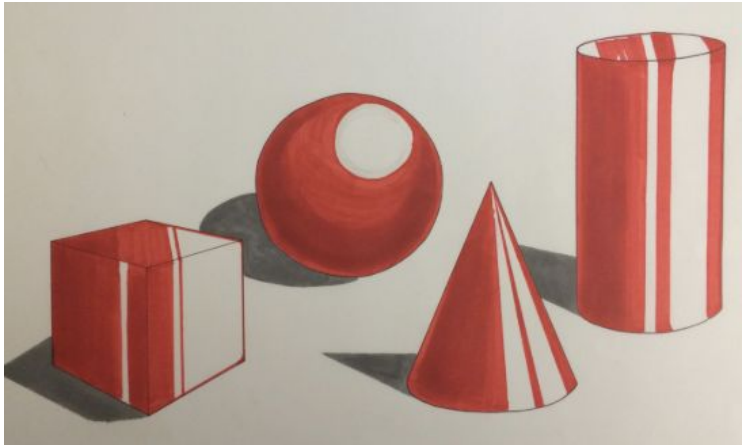
Learning Objectives:

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Task 2:

Use shading and rendering to complete your design.
Use annotation skills to stretch yourself and explain your design in detail.

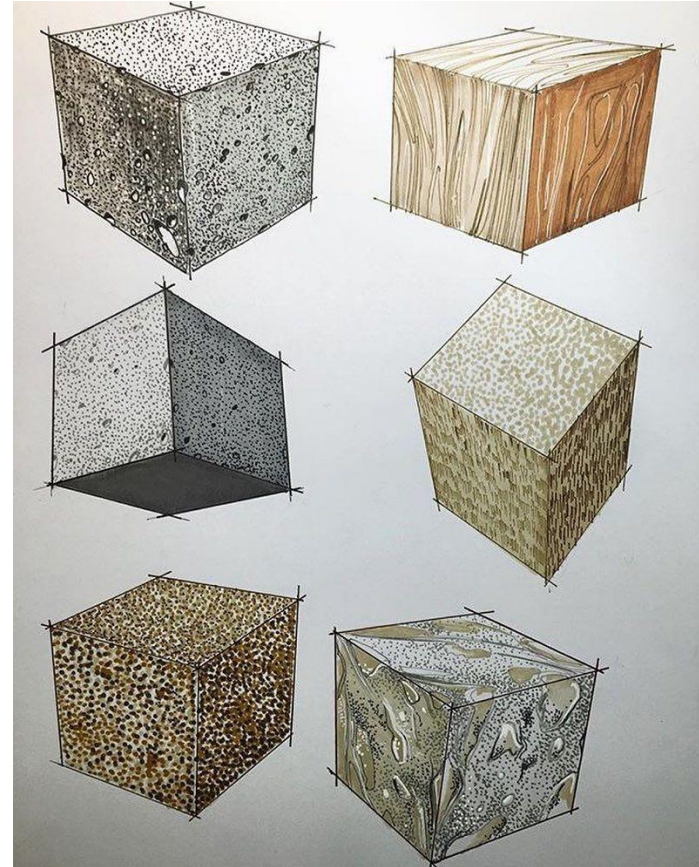


HELP HUB

Wood, Plastic,
Metal/Chrome, Glass

EXTENSION VIDEOS

How to draw textures with markers and colour pencils [Part 1](#)/[Part 2](#)





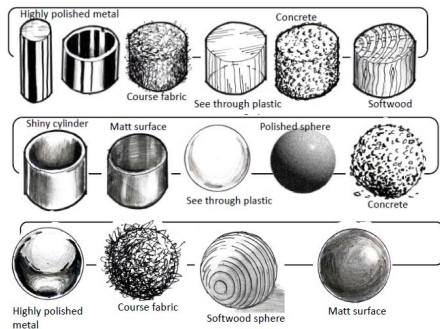
Rendering



What is rendering?

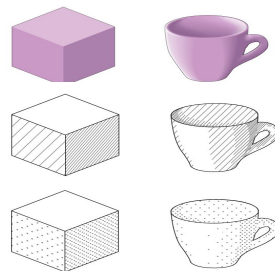
Creating the illusion of light, tone, and texture using graphic materials. Creating the illusion that an object is made from a particular material and describing how light falls on it.

Material effects

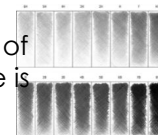


Wood grain
Cast metal
Glass
Plastic

Tone and shade



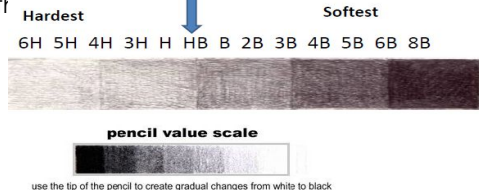
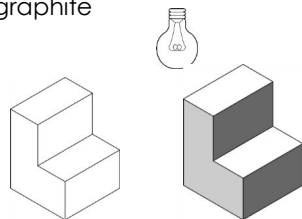
Tone refers to different **shades** of the same colour. A lighter tone is called a **tint** and is made by adding white to the base colour. A darker tone is called a **shade**.



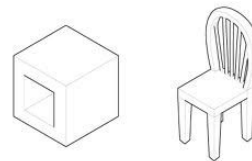
Tone can be shown by using shading, patterns of lines, or dots.

Using pencil shading to show tone. Pencil graphite can be hard or soft. The softer the graphite, the darker the tone it will make. The lightest tones are made with the hardest graphite.

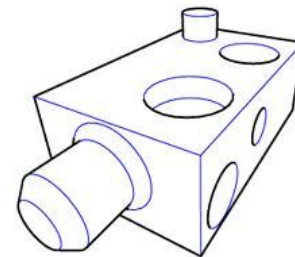
Light and shade



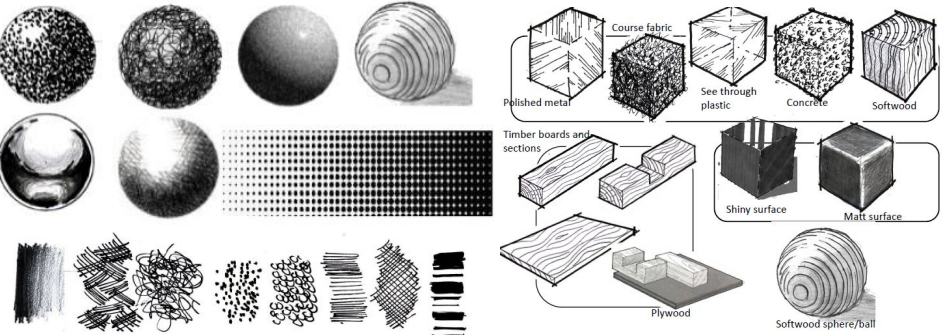
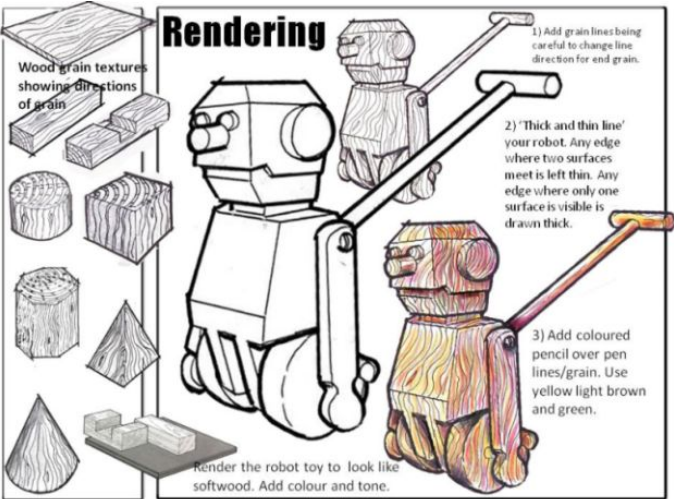
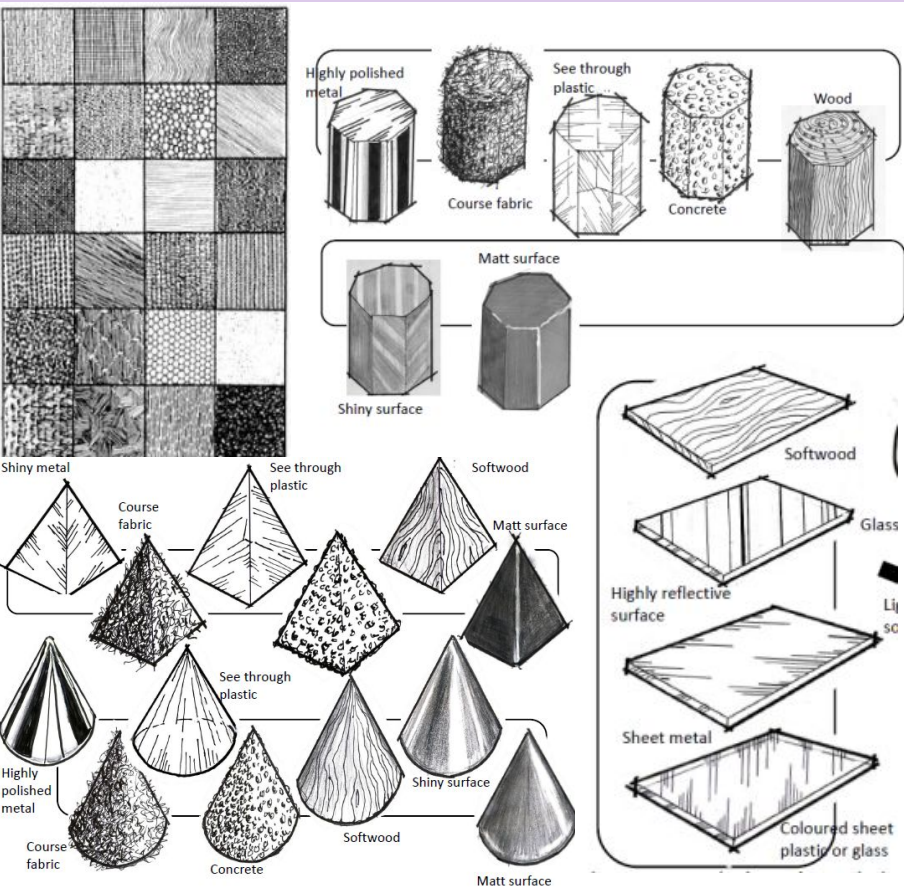
Thin lines show adjoining surfaces and thick lines are used when only one side can be seen.



Thick and thin lines



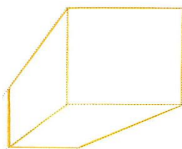
Rendering:
This means adding colour or texture to a picture. The aim of rendering is to make the drawing or sketch look more realistic, so you can get a good idea of what it looks like if it was to be made into a finished product. Some examples of rendering are; thick and thin line technique, shading & tone and material effects.



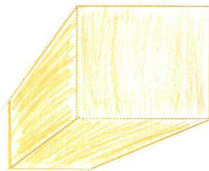
Rendering Help Sheet

Wood

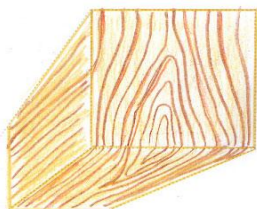
1) Draw outline of block using yellow or light brown.



2) Shade lightly, leaving white gaps.

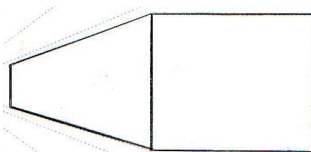


3) Add grain in a darker tone.

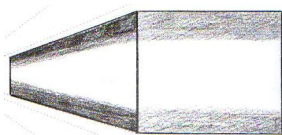


Metal

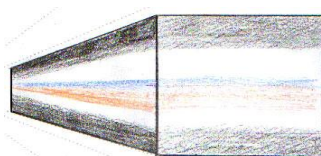
1) Draw outline of block lightly using a black pencil.



2) Shade in darker areas near the edges of each face, leaving the middle white.

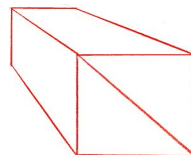


3) Add tints of colour in centre.

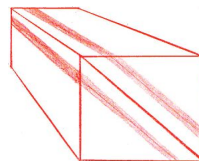


Plastic

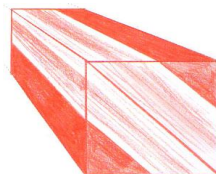
1) Pick a bright colour to draw in outlines and diagonal line on front face.



2) Add lighter line either side of the centre line and shade gently around it.

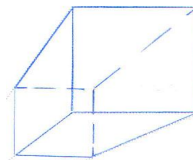


3) Leave small white gap and then shade darker towards the edges.

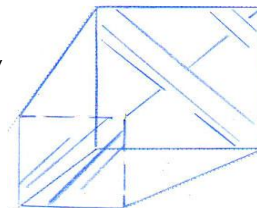


Glass

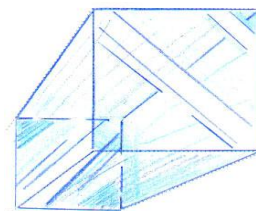
1) Use blue to draw outline, including lines you can see through the glass.

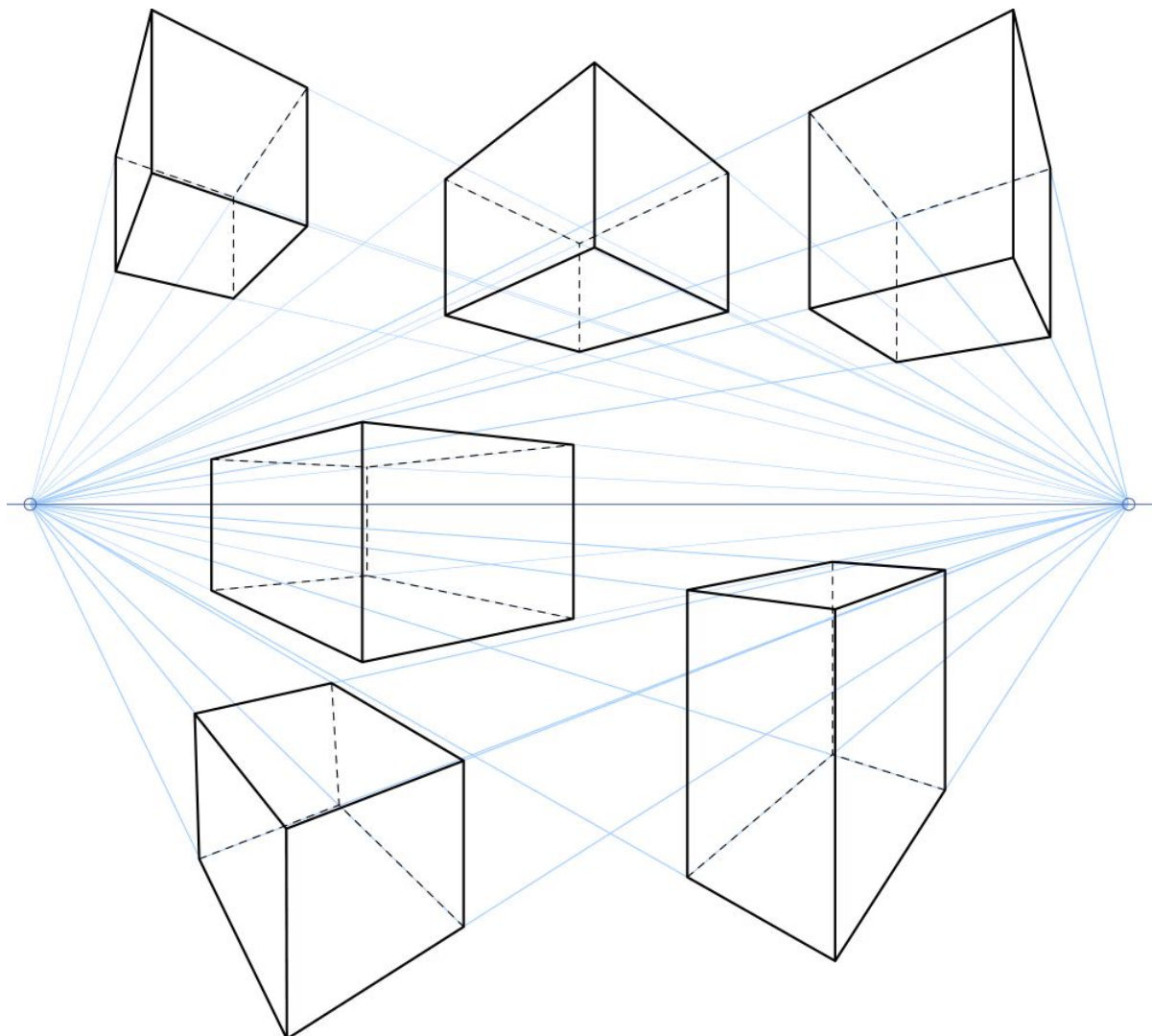


2) Draw diagonal lines in opposite directions across the front face and back face.



3) Shade lightly towards each edge of the glass.





Rendering Practice

Learning Objectives:

- Apply the **5Rs** when designing packaging.
- Develop **3 logo design ideas** that reflect their visual identity and target audience.
- Create **packaging design concepts** (front, back, and labelling).
- Produce **4–5 product design ideas**, including 3D rendering, dimensions, and annotations.

Think... Pair... Share...
Can you list the 5Rs?



Learning Objectives:

- Apply the **5Rs** when designing packaging.
- Develop **3 logo design ideas** that reflect their visual identity and target audience.
- Create **packaging design concepts** (front, back, and labelling).
- Produce **4–5 product design ideas**, including 3D rendering, dimensions, and annotations.



Learning Objectives:

- Apply the **5Rs** when designing packaging.
- Develop **3 logo design ideas** that reflect their visual identity and target audience.
- Create **packaging design concepts** (front, back, and labelling).
- Produce **4–5 product design ideas**, including 3D rendering, dimensions, and annotations.

Design Page 1:

1. Sketch 3 logo ideas that represent your brand or product identity. Explore typography, icons, and colour schemes.
2. Create their front and back packaging designs. Include brand name, logo, recycling info, product name, use instructions, and visual appeal.
- 3.

These can be done by hand or using a digital art platform such as procreate. Use websites such as dafont for typography ideas.

Design Page 2:

1. Sketch **4–5 product design ideas** in 3D.

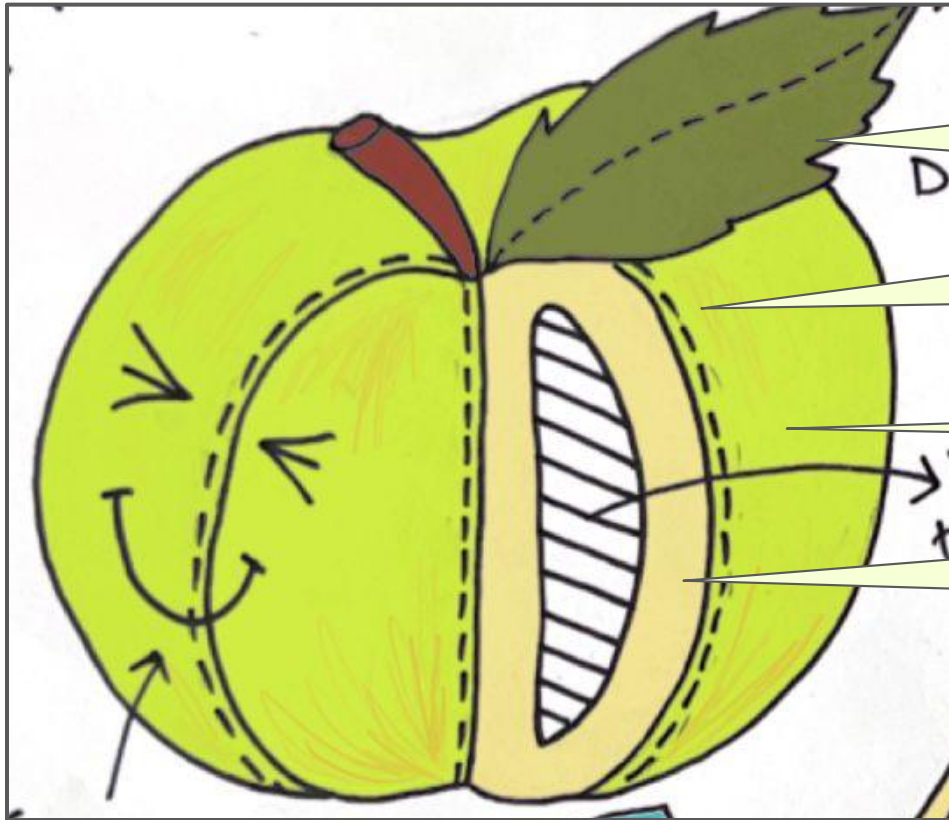
Each must include:

- ☐ Consistent design style
- ☐ Rendered appearance (colour/shading)
- ☐ Dimensions (approximate measurements)
- ☐ Annotations: *What did you like? What could be improved? How does it link to your situation/brief?*

You must include:

1. **3 Logo Designs** – hand-drawn or digital, exploring colour, style, and typography.
2. **Packaging Design (Front & Back)** – includes logo, product name, recycling info, user instructions, and brand style.
3. **Recycling Label** – use official symbols and clear information for disposal or reuse.
4. **4–5 Product Design Ideas** – drawn in 3D, rendered, annotated, and dimensioned.
5. **Annotations** – What works well? What could improve? How does it link to your situation and design brief?

Imagine you are a designer pitching this product to a client or producer.
What information would you need to include?



M...?

S...?

I...?

D... B...?

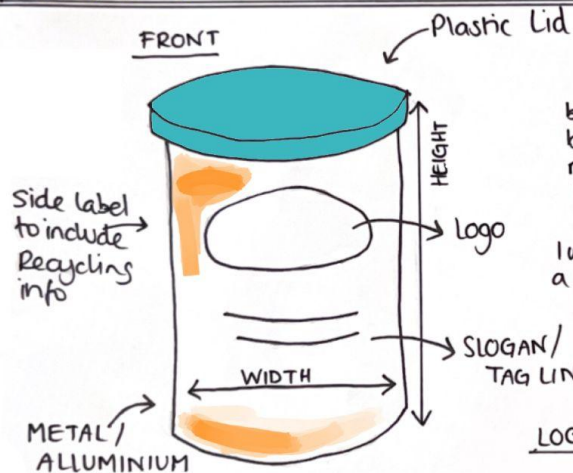
C...?

**Annotation
/标注/주석**

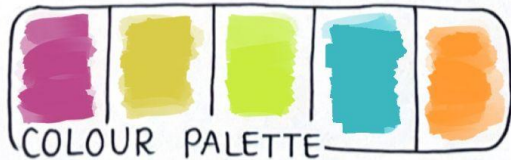
Brief notes that explain design ideas, materials, or choices on a drawing or model.

LOGO & PACKAGING DESIGN

WAGOLL



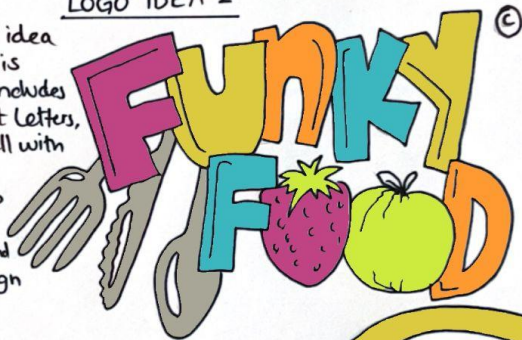
Blind box food plushy options. 1/9. Photos/realistic drawings



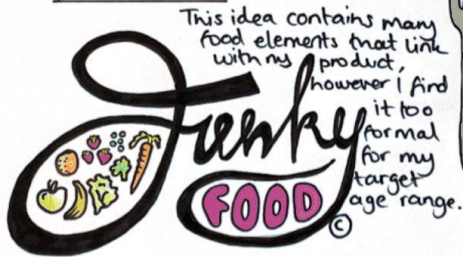
LOGO IDEA 2

I like this idea because it is bold and includes novelty fruit letters, it pairs well with my idea.

I will develop a border to better ground the design



LOGO IDEA 1

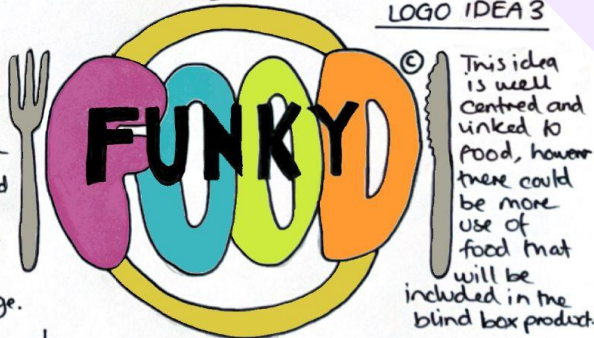


This idea contains many food elements that link with my product, however I find it too formal for my target age range.

SIDE LABEL

What recycling or reusing information could you add?

LOGO IDEA 3



This idea is well centred and linked to food, however there could be more use of food that will be included in the blind box product.

PRODUCT IN A TIN - PACKAGING IDEA

a brief paragraph of your packaging idea...

Information relating to the product 5Rs, any safety considerations, minimum age etc.

Annotation for each logo explaining why it related to your theme, what went well and what could be developed further.

An outline of the design ideas behind your packaging, why are you choosing certain colours, fonts, logo placement etc.

PRODUCT DESIGN IDEAS

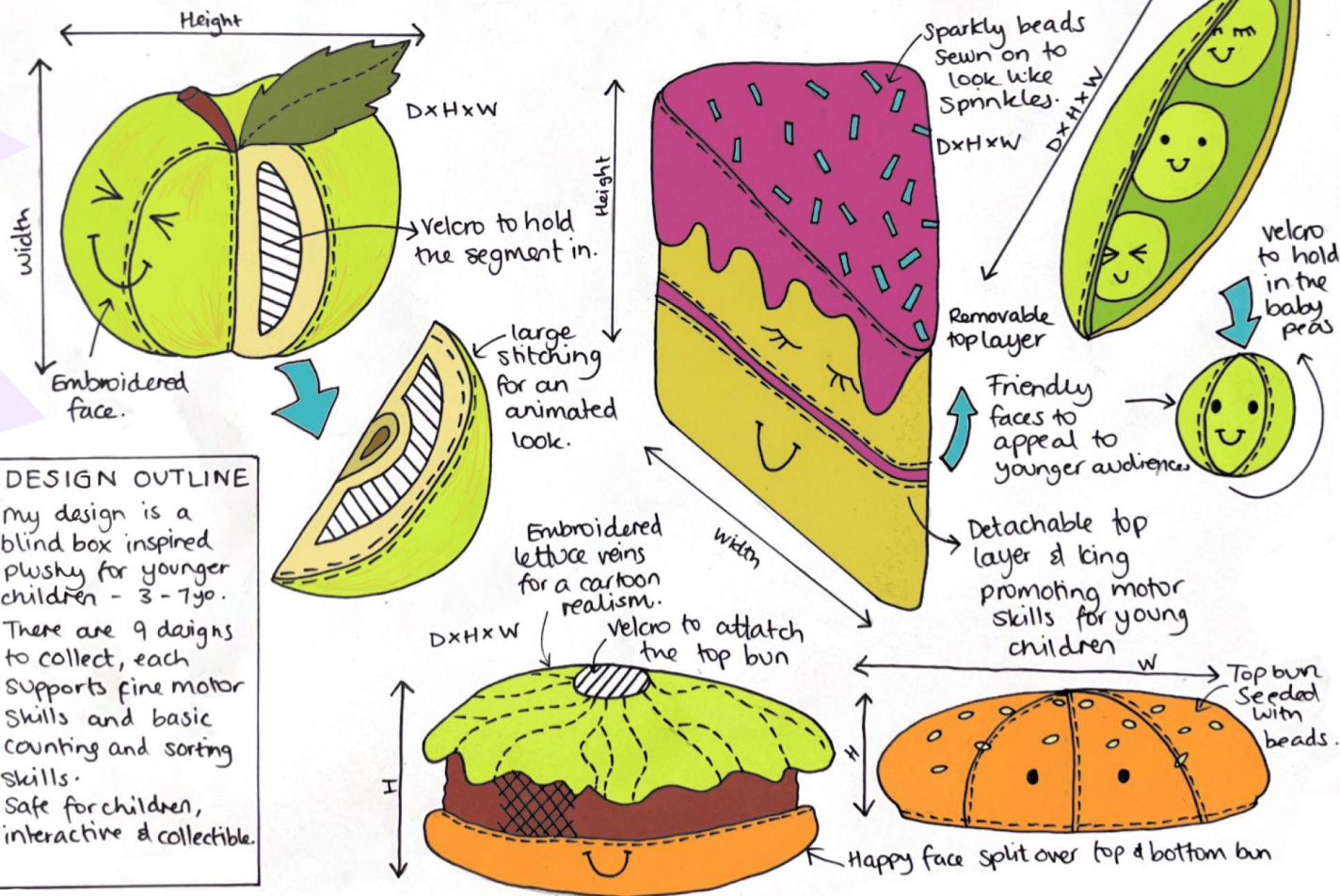
Detailed annotations about materials, size, fastenings, opening parts etc.

Design outline, explaining the ideas behind your designs, how it will work, who your customer is, why it links to your project etc.

Each design idea should be fully explained in annotations, well developed in concept linking to your theme and be rendered using colours, materials and textured finishes.

DESIGN OUTLINE

my design is a blind box inspired plushy for younger children - 3-7yo. There are 9 designs to collect, each supports fine motor skills and basic counting and sorting skills. Safe for children, interactive & collectible.



Learning Objectives:

- Work independently to manufacture their product safely and effectively.
- Apply quality control checks throughout the making process.
- Reflect on and record progress in a Making Diary, identifying challenges, mistakes, and solutions.
- Demonstrate problem-solving and self-guided decision-making skills.

Think, Pair, Share...

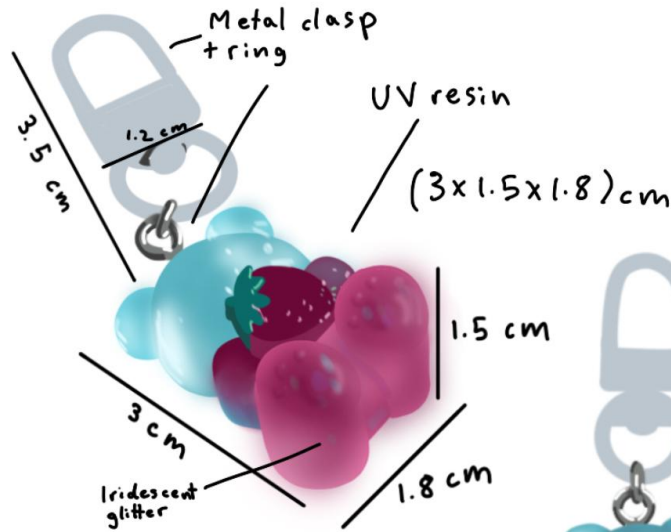
How was this product made?

How could we **evidence** how we make our products?

English Term	Mandarin	Korean	Meaning / Use
Making / Manufacture	制作 (zhì zuò)	제작 (jejag)	The process of constructing or building your product.
Evidence	证据 (zhèng jù)	증거 (jeung-geo)	Proof of progress or learning (e.g., photos, notes, sketches).
Mistake / Error	错误 (cuò wù)	실수 (silsu)	Something done incorrectly that needs fixing.



Product design ideas

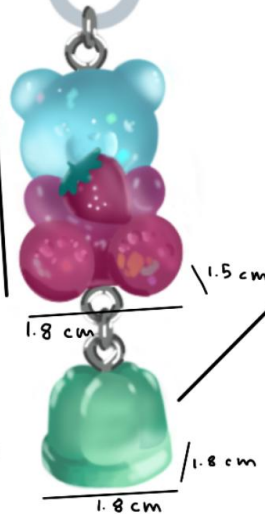


Design Outline

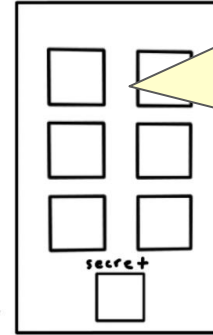
My design is a blind box inspired keychain aimed at 12~20 yo females. There will be a total of 6 designs to collect with a secret design (1/10). Each design will have its own pastel colour scheme. Cute, colourful and addictive.

3 cm
(3 x 1.8 x 1.5)

(1.2 x 1.8 x 1.8)



Each tin comes with its own sheet showing the designs available. Each design will be shown in a simple, illustrated style.



AO1

Is it clear who this product is designed for?

AO2

Is this design page well researched and clear? Why or why not?

AO3

Does this design page show good drawing skills and annotation?

AO	Focus	Emerging	Developing	Secure	Extending
AO1	Identifying Need & Design Brief <i>Why the design is needed & who for</i>	I can describe an idea but I don't clearly explain why it's needed or who it's for.	I can identify a need or problem and state who might use my product.	I can clearly explain why my design is needed, who it is for, and what it should do.	I have analysed a real design opportunity and written a focused, justified brief that shows why, who and what my product must achieve.
AO2	Research & Specification <i>Gathering info and making goals</i>	I have collected a small amount of information, but it isn't well linked to my project.	I have gathered some relevant research and used it to make early design choices.	I have researched carefully and used my findings to create a clear, realistic specification.	I have carried out detailed, comparative research and produced an evidence-based specification that clearly defines measurable requirements for my product.
AO3	Generating & Exploring Ideas <i>Sketching, 3D ideas, annotations</i>	I have drawn a few basic ideas but given little explanation or variety.	I have produced several different ideas and explained some features with notes.	I have produced a range of creative ideas with clear sketches/annotations that meet my specification.	I have generated a wide and original variety of well-presented ideas, using strong visual language to show innovation and clear links to the specification.

Name: _____

Date: _____

Checklist:

- ☐ **Page 1:** Title Page
- ☐ **Page 2:** Product in a Tin
Mindmap
- ☐ **Page 3:** Situation &
Design Brief
- ☐ **Page 4:** Customer Profile
- ☐ **Page 5:** 2 Point
Perspective
- ☐ **Page 6:** Logo and
Packaging Design
- ☐ **Page 7:** Product Design
Ideas

Learning Objectives:

- Work independently to manufacture their product safely and effectively.
- Apply quality control checks throughout the making process.
- Reflect on and record progress in a Making Diary, identifying challenges, mistakes, and solutions.
- Demonstrate problem-solving and self-guided decision-making skills.

How to make progress...	
All	Work safely, follow plan, and record making progress.
Most	Identify and fix small errors independently, showing evidence in diary and product.
Some	Demonstrate high-quality finish and reflective diary with photos, problem-solving notes, and clear improvements.

Checkpoint	What to Look For	Action if Problem Found
Accuracy	Are parts measured and cut correctly?	Re-measure, trim, adjust or remake small components.
Assembly	Do pieces fit together properly?	Check alignment, rejoin using appropriate method.
Finish	Is the surface smooth, clean, or painted evenly?	Sand, repaint, or reapply finish carefully.
Function	Does it fit inside the tin and work as intended?	Redesign small elements, test and adjust.
Safety	Are sharp edges or hazards present?	Smooth, seal, or reinforce as needed.

Task:

To begin or continue constructing your product.

Expectations During Making

You should:

- Consider your plan for making, materials, equipment and the safety precautions.
- Record process, decisions, and changes in their Making Diary (can be handwritten or digital).
- Take photos or sketches at key stages (for evidence of process).
- Conduct Quality Control checks every 15–20 mins using the checklist below.

Learning Objectives:

- Work independently to manufacture their product safely and effectively.
- Apply quality control checks throughout the making process.
- Reflect on and record progress in a Making Diary, identifying challenges, mistakes, and solutions.
- Demonstrate problem-solving and self-guided decision-making skills.

English Term	Mandarin	Korean	Meaning / Use
Making / Manufacture	制作 (zhì zuò)	제작 (jejag)	The process of constructing or building your product.
Material	材料 (cái liào)	재료 (jaeryo)	The substance used to make your product.
Tool	工具 (gōng jù)	도구 (dogu)	The equipment used in making or shaping your product.
Accuracy	精确 (jīng què)	정확성 (jeonghwakseong)	How exact or precise your measurements and work are.
Function	功能 (gōng néng)	기능 (gineung)	What the product is meant to do.
Quality Control	质量控制 (zhì liàng kòng zhì)	품질 관리 (pumjil gwanri)	Checking that your product meets a good standard.
Mistake / Error	错误 (cuò wù)	실수 (silsu)	Something done incorrectly that needs fixing.
Fix / Repair	修复 (xiū fù)	수리하다 (surihada)	To correct or mend something that broke or failed.
Evaluate	评估 (píng gū)	평가하다 (pyeonggahada)	To judge how successful your product or process is.
Reflection	反思 (fǎn sī)	성찰 (seongchal)	Thinking about what went well and what can improve.

Diary of Making

Burger Bottom Edge
Cut x1

Burger Top
Cut x1

Burger
Top/Bottom
Bun
Cut x3

Burger
Top/Bottom
Bun
Cut x3

Burger
Top/Bottom
Bun
Cut x3

Step 1: I drew the shapes out that I required for my food items. I used a stiff felt fabric which doesn't fray, is easy to sew and soft for use.

Step 2: I used basic backstitch to sew together the burger bottom pieces to the edge piece.

Burger
Patty
Outer
Cut x2

Burger
Patty
Inner
Cut x4

Burger
Patty
Inner
Cut x4

Burger
Patty
Outer
Cut x2

Burger
Patty
Inner
Cut x4

Burger
Patty
Inner
Cut x4



Step 7: I used backstitch to sew the top and bottom burger patty together.

Step 8: I stacked the 4 inner burger patties and padded the burger out. Then I sewed the edge and added chain embroidery for griddle lines.



Step 3: I used backstitch to sew the points together to form the dome of the top bun, I then added yellow sequins to represent seeds.

Step 4: I stuffed the top and bottom bun with polyester filling and stitched them closed.

Step 5: I used a mix of backstitch and blanket stitch to carefully assemble the tomato, the pink inner sandwiched between the two outer tomato pieces.

Cheese
Slice Cut
x2

Step 9: I sandwiched the cheese sliced together and joined with blanket stitch.

Lettuce
Leaf
Cut x1

Tomato
Inner
Cut x1

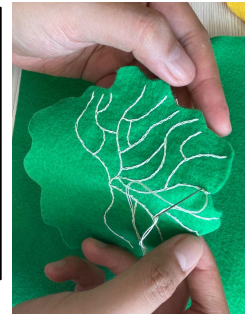
Tomato
Outer
Cut x2



Step 6: I used small white beads as the tomato seeds, I attached with thin white thread carefully so none could be seen on the pink background.



Step 10: The lettuce veins were sewn using fine thread to create a chain stitch, the lettuce is a single layer.

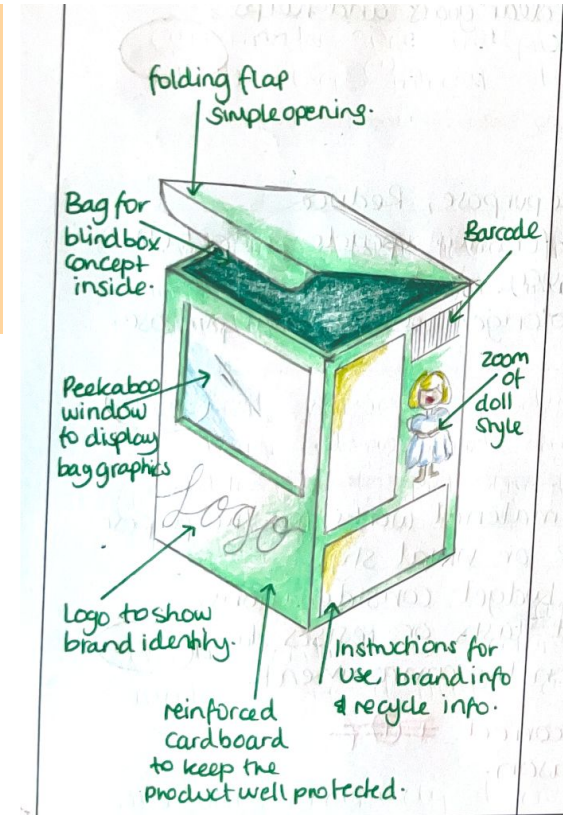
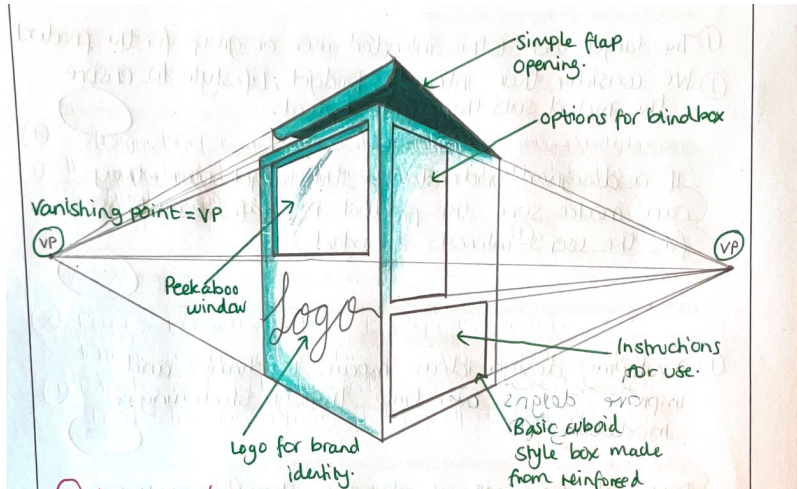


Learning Objectives

- I can identify my strengths and areas for development by analysing my common assessment results.
- I can use the GROW reflection model to set clear, achievable next steps for improving my Design & Technology skills.

Think Pair Share:

1. What part of the assessment did you feel most confident about?
2. What part felt the most challenging?



Learning Objectives

- I can identify my strengths and areas for development by analysing my common assessment results.
- I can use the GROW reflection model to set clear, achievable next steps for improving my Design & Technology skills.

- Use **green pen** to develop your answers using the example assessment.
- Reflect on your work in green pen by answering **ALL** of the following questions on the back of your paper:
 - Where did I gain the most marks? Why?
 - Where did I lose marks? What was missing or unclear?
 - What could I do to improve my design sketches?
 - Which skills can I practice more often?
 - What are the top 2-3 things I will do differently next time?
 - What will I change immediately in class?
 - How will I measure my improvement?

11/20/25, 11:42 AM Design and Technology Common Assessment Y9 Term1b - Google Docs

50/50

Design and Technology Common Assessment Y9 Term1b

Section A – Design Understanding (10 marks)

This section tests your knowledge of key design concepts.

Q1. What is a design brief, and why is it important when starting a project? (2 marks)

① A design brief is a statement of intent for your design.
 ① It's important because it gives clear goals and helps to guide design choices.

Q2. Describe what the 5Rs of sustainability stand for. Explain one example of how you have applied one of them in your project. (5 marks)

③ Reuse, Recycle, Refuse, Repurpose, Reduce
 I have repurposed a tin for my upcycle project ①
 ① which will help to reduce waste and ensure that the tin life is prolonged with a new purpose.

Q3. Name three factors that can influence a designer's choice of materials. Explain each briefly. (3 marks)

① Function - how well the material works for the purpose
 ① Aesthetics - colour, texture or visual style
 ① Cost - Affordability or budget considerations.
 Durability - how long it lasts or resists damage.
 sustainability - impact on the environment.

* ① mark for each correct ~~FF~~ including the reason.

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11/20/25, 11:42 AM Design and Technology Common Assessment Y9 Term1b - Google Docs

Section B – Design Ideas & Development (15 marks)

This section assesses your ability to communicate and develop design ideas visually.

Q4. The original brief asked you to create a product that fits inside a tin. Sketch **two alternative packaging ideas** that could replace the tin but still protect and present your product attractively. Each sketch should be:

- neat and labelled
- annotated with materials, colours, and purpose

A different design here.

- different shape
- different colours
- different construction.

① ~~Neat~~ Neat design
 ① Labels & clear annotation
 ① Colours & purpose clear.

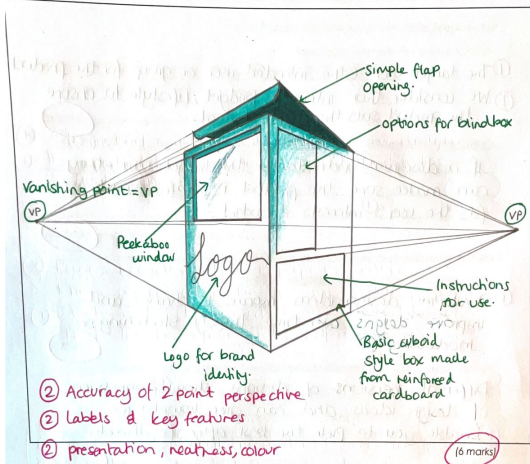
(8 marks) 3 marks per idea

https://docs.google.com/document/d/1AoCfndN1_ymSp2wudPwlog7CKGdQyQK0xTQ/edit#heading=h.7y92mhw 2/5

Learning Objectives

- I can identify my strengths and areas for development by analysing my common assessment results.
- I can use the GROW reflection model to set clear, achievable next steps for improving my Design & Technology skills.

Q5. Choose one of your packaging ideas and draw it using 2-point perspective to show shape and depth. Add simple labels for key features.



Q4. Describe how your packaging design helps communicate the product's purpose or brand identity.

- The logo & colour/imagery represent a clear brand identity.
 - The effective use of bright colours helps my design appeal to the target user which is children. The design & imagery show a playful style.
- (3 marks)

Section C – Design Thinking & Application (15 marks)

This section tests how well you understand and apply the design process.

Q7. Designers often think about who their product is for.

a) What is meant by the term target user?

- The target user is the intended user or group for the product.
 - We consider their interests, budget, lifestyle to ensure the product suits their wants & needs.
- (2 marks)

b) Explain why it is important to understand the target user when designing.

- If a designer understands the target user, they can make sure the product is safe, easy to use, fits the user's interests & budget.
- (3 marks)

Q8. During the design process, ideas often change and improve.

a) Describe what is meant by developing ideas in design.

- Developing design ideas means to change and improve designs over time through sketching or modelling.
- (2 marks)

b) Give one reason why it is important to show different versions of your design ideas.

- Different versions of designs show progression of design ideas and can give variety to enable you to pick the best after feedback.
- (3 marks)

Q9. Quality control helps designers check their work.

a) Define quality control.

- Quality control is checking your work while making sure it meets a good standard of making.
- (2 marks)

b) Give one example of a quality control check that could be used when making a product.

- Measuring accurately & cross referencing to designs
- (1 mark)

to ensure accuracy before cutting and to avoid waste materials.

(3 marks)

Section D – Reflection & Evaluation (10 marks)

This section helps you reflect on your progress and understanding.

Q10. Explain why evaluation is important throughout the design process, not just at the end.

- Evaluation helps designers see problems earlier and make improvements, leading to a better final product.
- (3 marks)

Q11. Designers learn from mistakes and feedback.

a) Describe a situation where a mistake during making could lead to a design improvement.

- If the lid to my tin was not secure or poorly fitted, I could redesign a more secure version or a style with a clip or secure fastening.
- (3 marks)

b) Suggest two ways a designer can record and reflect on their progress during a project.

- Diary of making: recording progress with photos & videos to track the making of the product.
 - User feedback: collect user feedback and show clear *photos, videos, notes* reflection of how the product meets user needs & how you are developing.
- (4 marks)

1 mark per style of recording.

1 mark for explanation of style.

Learning Objectives:

- Work independently to manufacture their product safely and effectively.
- Apply quality control checks throughout the making process.
- Reflect on and record progress in a Making Diary, identifying challenges, mistakes, and solutions.
- Demonstrate problem-solving and self-guided decision-making skills.

Evaluation Questions:

1. **How well does your final product meet your design brief and the needs of your target user?**

→ (Does it work as you planned? Would your customer actually want or use it?)

2. **What went well during the making process and what skills did you improve the most?**

→ (Think about tools, materials, accuracy, or creativity.)

3. **What problems or setbacks did you face, and how did you overcome them?**

→ (Be specific — what did you do when something went wrong?)

4. **How does your packaging and branding show the values of the 5Rs (Reduce, Reuse, Recycle, Refuse, Rethink)?**

→ (Did you design it to be more sustainable or reusable?)

5. **If you could improve one part of your design or product, what would it be and why?**

→ (Be honest — what would make it more functional, attractive, or durable?)

6. **What did you learn about being an independent designer and maker from this project?**

→ (How have your confidence, creativity, or problem-solving skills developed?)

How to make progress...	
All	Describe what went well and what could be improved in their product.
Most	Explain how their final product meets their design brief and how they solved problems during making.
Some	Critically evaluate their product and packaging, linking reflections to the 5Rs, design decisions, and target user feedback.

Materials & Equipment

- Felt (Green, Orange, Brown, Yellow, Red, Pink)
- Embroidery Thread
- Fine Thread
- Yellow Sequins
- White Beads
- Polyester Stuffing
- Embroidery Needles
- Fine Thread Needles
- Scissors
- Pinking Shears

*Photos of your
materials and
equipment
here*

Front Final Photo

Of your product and packaging (tin)

Back Final Photo

Of your product and packaging (tin)

Diary of Making

Evaluation

1. How well does your final product meet your design brief and the needs of your target user?
2. What went well during the making process and what skills did you improve the most?
3. What problems or setbacks did you face, and how did you overcome them?
4. How does your packaging and branding show the values of the 5Rs (Reduce, Reuse, Recycle, Refuse, Rethink)?
5. If you could improve one part of your design or product, what would it be and why?
6. What did you learn about being an independent designer and maker from this project?

Name: _____

Date: _____

AO	Focus	Emerging	Developing	Secure	Extending
AO4	Development of Proposed Solution <i>Modelling, testing, refinement</i>	I have shown limited development or testing of my ideas.	I have started to test and improve ideas using feedback.	I have developed and refined my ideas based on testing and feedback, producing a clear final solution.	I have iteratively tested and refined prototypes, used evidence and feedback to justify design decisions, and documented substantial development toward an improved final solution.
AO5	Planning for Production <i>Steps, tools, materials, drawings</i>	I have a basic plan but it lacks clear steps, tools or materials.	I have planned main stages and listed likely tools and materials.	I have produced a logical, step-by-step plan with materials, tools and basic working dimensions.	I have created a professional production plan with accurate working drawings, full material/tool lists, safety checks and contingencies to prevent problems.
AO6	Product Realisation (Making) <i>Practical skill, accuracy, finish</i>	I have needed a lot of help and my product is incomplete or inconsistent.	I have made a working product but it has some inaccuracies or finish issues.	I have made a functional, well-finished product using tools safely and showing good accuracy.	I have produced an outstanding, precise product showing excellent craftsmanship, independence and creative skill; my finish and function exceed expectations.
AO7	Testing & Evaluation <i>Evidence-based reflection & improvements</i>	I have given simple opinions with little evidence or link to my brief.	I have tested my product and identified some things that worked and didn't.	I have evaluated my product clearly, using evidence and feedback to suggest realistic improvements.	I have produced a detailed, evidence-based evaluation, interpreting test/feedback results to justify strengths and propose innovative, realistic next-steps for development.

Checklist:

- ☐ **Page 8:** Final Design Idea
- ☐ **Page 9-10:** Diary of making, final photos and Evaluation
- ☐ **Page 11:** Peer assessment feedback (only done at final presentation)
- ☐ **Page 12+:** Common assessment scans with Green Pen development of answers and reflection questions answered on the back (6 pages total)
- ☐ **+ Your final physical product**
- ☐ **+ Your final physical tin/packaging**

Learning Objectives:

- Work independently to manufacture their product safely and effectively.
- Apply quality control checks throughout the making process.
- Reflect on and record progress in a Making Diary, identifying challenges, mistakes, and solutions.
- Demonstrate problem-solving and self-guided decision-making skills.

Task:

1. Display your finished tins and products and printed portfolios.
2. Complete a 'Gallery Walk' get a good look at everyone's tins.

I will assign you one tin to peer asses.

3. Use the peer feedback notes provided to give peer feedback: two "WWW" (what works well) and one "EBI" (even better if... or what could be improved).

Keywords: design, customer, function, function, accuracy, aesthetics, sustainability.

PEER Feedback		
Peer Assessor: _____		Date: _____
WWW	WWW	EBI

Write in FULL SENTENCES

**Things to consider...**

- How well have they linked their final product and packaging to their situation and design brief?
- Does their final product and packaging suit the wants and needs of their customer/target market?
- Do the overall aesthetics of their final product and packlink to the aesthetics used in their portfolio?
- Is their final product and packaging well finished?

PEER Feedback

Peer Assessor: _____
Date: _____

WWW

WWW

EBI

PEER Feedback

Peer Assessor: _____
Date: _____

WWW

WWW

EBI

PEER Feedback

Peer Assessor: _____
Date: _____

WWW

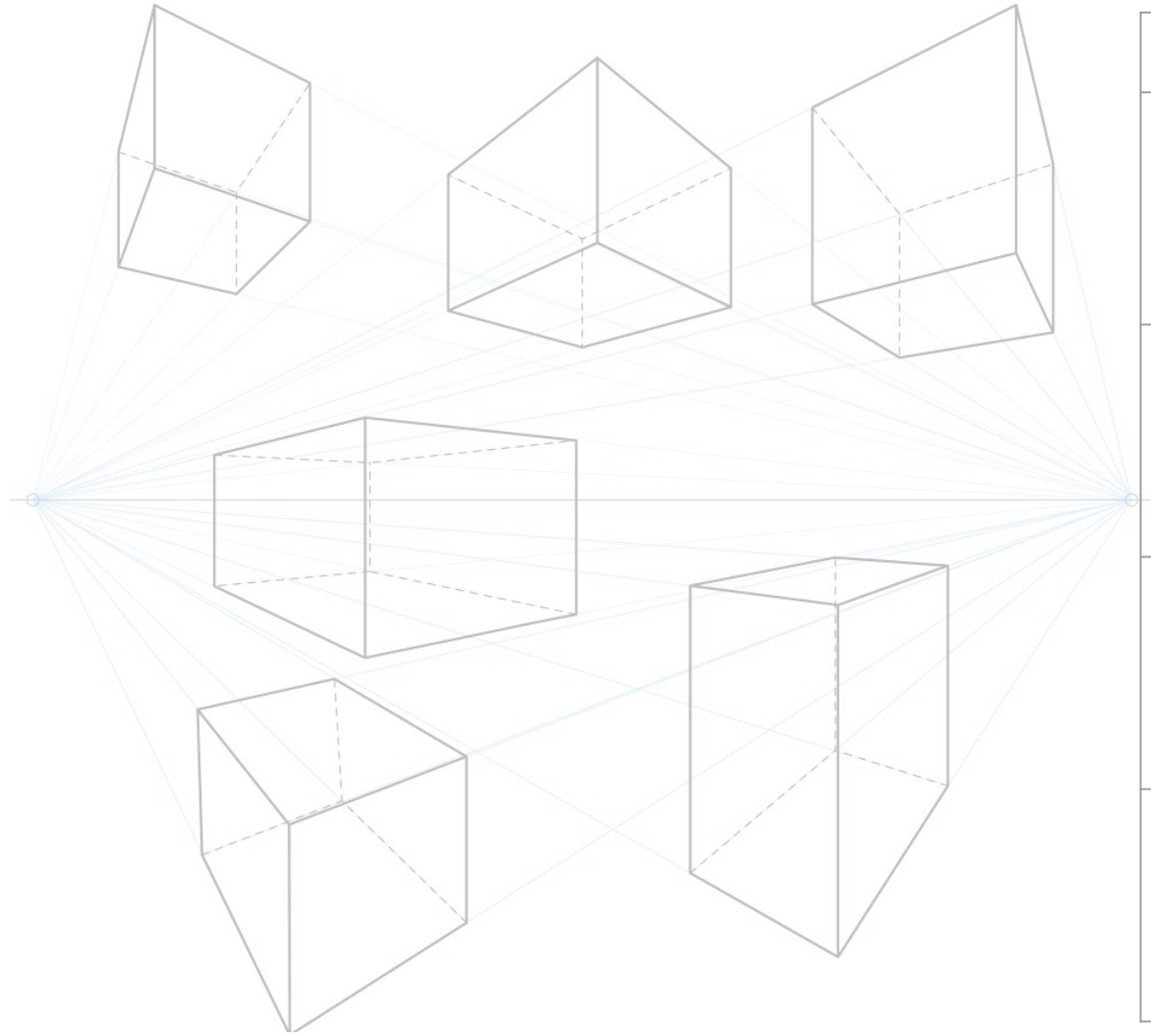
WWW

EBI

AO	Focus	Below	Meeting	Exceeding	Mastering
AO1	Identifying Need & Design Brief <i>Why the design is needed & who for</i>	I can describe an idea but I don't clearly explain why it's needed or who it's for.	I can identify a need or problem and state who might use my product.	I can clearly explain why my design is needed, who it is for, and what it should do.	I have analysed a real design opportunity and written a focused, justified brief that shows why, who and what my product must achieve.
AO2	Research & Specification <i>Gathering info and making goals</i>	I have collected a small amount of information, but it isn't well linked to my project.	I have gathered some relevant research and used it to make early design choices.	I have researched carefully and used my findings to create a clear, realistic specification.	I have carried out detailed, comparative research and produced an evidence-based specification that clearly defines measurable requirements for my product.
AO3	Generating & Exploring Ideas <i>Sketching, 3D ideas, annotations</i>	I have drawn a few basic ideas but given little explanation or variety.	I have produced several different ideas and explained some features with notes.	I have produced a range of creative ideas with clear sketches/annotations that meet my specification.	I have generated a wide and original variety of well-presented ideas, using strong visual language to show innovation and clear links to the specification.
AO4	Development of Proposed Solution <i>Modelling, testing, refinement</i>	I have shown limited development or testing of my ideas.	I have started to test and improve ideas using feedback.	I have developed and refined my ideas based on testing and feedback, producing a clear final solution.	I have iteratively tested and refined prototypes, used evidence and feedback to justify design decisions, and documented substantial development toward an improved final solution.
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ACCESSFM

	A	Aesthetics	Aesthetics means what does the product look like? What is the Colour? Shape? Texture? Pattern? Appearance? Feel? Weight? Style?
	C	Cost	Cost means how much does the product cost to buy? How much does the product cost to make? How much do the different materials cost? Is it worth it?
	C	Client	Client /Customer/User means who will buy your product? What is their Age? Gender? Income? Lifestyle? User needs? Likes/Dislikes? Needs? Preferences?
	E	Environment	Environment means will the product affect the environment? Is the product; recyclable? Reusable? Repairable? Sustainable? Environmentally friendly? BAAd for the environment? 6R's of design: Recycle, Reuse, Repair, Rethink, Reduce, Refuse.
	S	Size	Size means how big or small the product is. What is the size of the product in millimeters (mm)? Is the size similar to other products on the market? Is it comfortable to use? Does it fit? Would it be improved if it was bigger or smaller?
	S	Safety	Safety means how safe is the product when it is in use? Will it be safe for the customer to use? Could they hurt themselves? What is the correct and safest way to use the product?
	F	Function	Function means how does the product work? What is the product's job and role? What is it needed for? How well does it work? How could it be improved? Why is it used this way?
	M	Materials	Materials means what is the product made from? Why were these materials used? Would a different material be better? How was the product made? What manufacturing techniques were used?



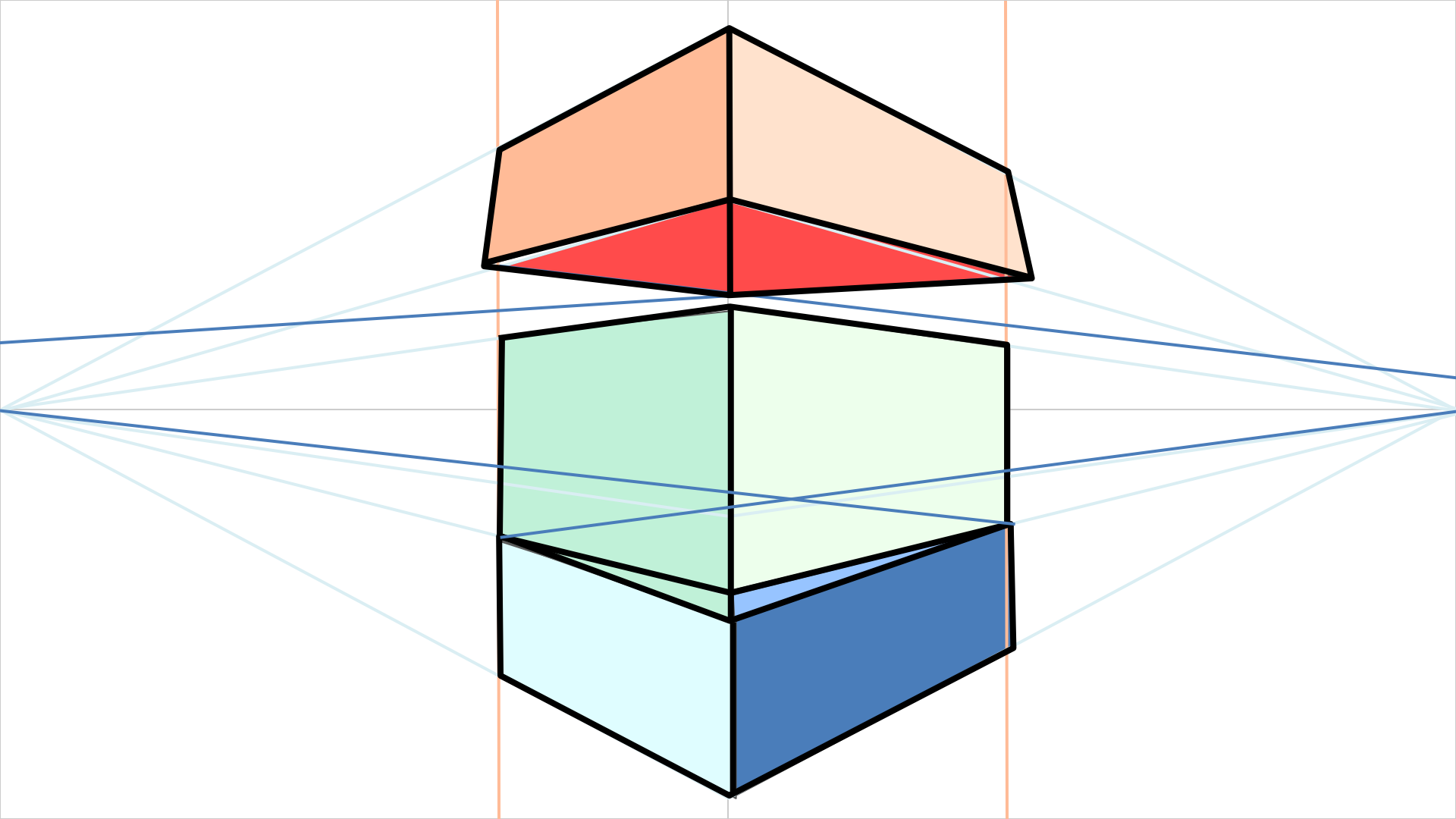
Rendering Practice

Wood

Metal

Plastic

Glass



Assessment Objectives Projects																
	AO1															
	AO2															
	AO3															
	AO4															

Excellent		Secure		Developing		Foundation	
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Assessment Objectives Projects																
	AO1															
	AO2															
	AO3															
	AO4															

Excellent		Secure		Developing		Foundation	
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